

CRISIS

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EDITOR'S NOTE



Yana Petri



Aarohi Bhargava-Shah

Crisis. In many ways, the theme of this issue evokes a sentiment that has been particularly salient in our community during the past few months. The 2018 wildfires that ravaged several parts of California for days on end—and personally affected many students right here on campus—were some of the most destructive and deadly on record. On a national scale, the continuing opioid addiction epidemic constitutes one of our nation's foremost public health crises. And, of course, the sustained political moment in which we have found ourselves of late has left citizens of all political persuasions feeling distrustful and disillusioned by our leaders.

But while these crises vary in magnitude and scope, they each, in their own way, have either been grounded in scientific phenomena or made an indelible impact on scientific discourse. In this issue, our dedicated team of writers and editors seeks to explore this intricate relationship between scientific progress and these critical moments in our society, by tackling a wide range of issues. What, for example, are the ecological and economic impacts of constructing dams in areas of great biodiversity? How are leading experts such as Professor Eva Harris, Director of the Center for Global Public Health at Berkeley, leading the charge in tackling massive global health epidemics such as Zika and dengue? Taking a look inward, how might scientists alter the way they communicate with the public to combat misinformation with facts? As ever, we find that even the most intractable challenges have solutions that lie in the fascinating discoveries being made by scientists all over the world every day.

The *Berkeley Scientific Journal* is proud to report that it continues to uphold its commitment to responsible science journalism while making innovative strides in its mission of improving scientific literacy and training students in producing clear written and visual communication. For the first time ever, our team of editors hosted a hands-on science demonstration at the Bay Area Science Festival, which garnered an enthusiastic response. And with the continued support of our community of readers, we raised over \$3000 this year through our Crowdfunding campaign, allowing us to pursue even more exciting projects such as the publication of this journal in print.

With these bright prospects ahead, we are thrilled to present this thought-provoking issue of the *Berkeley Scientific Journal*.

Aarohi Bhargava-Shah
Managing Editor

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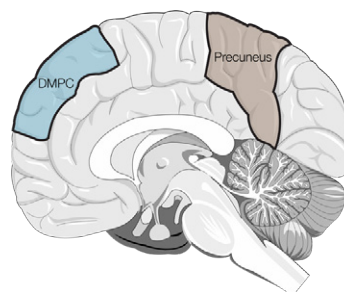
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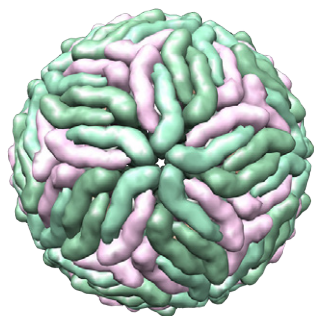
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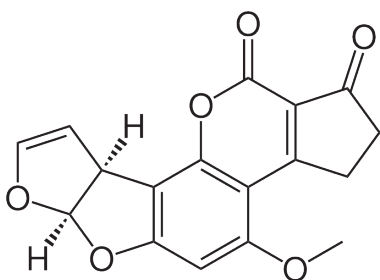
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THE SCIENCE OF SCIENCE RHETORIC

BY JONATHAN KUO

One of the earliest collections of writing samples in human history comes from what is often called the “birthplace of civilization”: ancient Mesopotamia. Containing deep indentations carved into hardened clay, Mesopotamian tablets encapsulate a wealth of information ranging from the minutiae of daily life to elaborate myths such as the Epic of Gilgamesh. It should come as no surprise, then, that Mesopotamia is one of several starting points for rhetoric. Indeed, the oldest known letter of complaint originates from Mesopotamia and describes an argument between a copper ore merchant and an unhappy customer:

You alone treat my messenger with contempt! [...] Take cognizance that (from now on) I will not accept here any copper from you that is not of fine quality [...] and I shall exercise against you my right of rejection because you have treated me with contempt.¹

But how do we operationally define—or even define—rhetoric in a scientific setting? In natural science, operational definitions precisely delineate how quantities are measured. One operational definition of sleep, for instance, measures the pattern of

EEG waveforms observed during a period of unconsciousness. The primary issue that arises in creating an operational definition of rhetoric, however, is that of scope. Rhetoric is typically thought of as the art of persuasion. Formal rhetoricians seek to

convince, persuade, and at times manipulate their audience into accepting a particular argument. On the other hand, persuasion is not necessarily confined to such oratorical discourse. The trope of a boy pulling a girl’s hair in school, a paper published in a



► *Figure 1: One of many Mesopotamian tablets currently housed at the Walters Art Museum in Baltimore, Maryland.*

scientific journal, and heated debates over political events in Facebook comments are all modern examples of persuasion that do not necessarily take the form of speech. Forms of rhetoric have, of course, already been defined in academic literature. A research review in *Discourse Processes* compiled several definitions of "argumentation" as based on making a concept more accessible, justifying an uncertain position, or improving an audience's understanding of an idea.² But for now, let's focus on exploring the parts of rhetoric that try to change psychological cognition with the aim of conveying certain ideas.

IN SEARCH OF QUANTIFICATION

The study of rhetoric from a cognitive neuroscience perspective is most frequently operationalized using functional magnetic resonance imaging (fMRI). The basic premise of fMRI is that the activation of neuronal circuits is accompanied by greater flow of oxygenated blood to those areas. Because oxygenated blood has different magnetic properties than deoxygenated blood, fMRI machines can detect activated brain regions in real time.³

Since the invention of fMRI in 1990, researchers have used the technique to map a variety of brain structures to their corresponding functions, although many of these remain incompletely characterized. In a study at the University of Michigan, researchers used fMRI to examine parts of the brain associated with processing of self-relevant messages (messages that are perceived as relevant to one's self) and smoking cessation messages. The scientists hypothesized that since personalized treatment plans increase rates of successfully quitting smoking, these plans should activate neural regions associated with smoking-cessation and self-relevant messages. Furthermore, the researchers collected fMRI data to predict whether a smoker given a treatment plan would successfully quit based on their level of

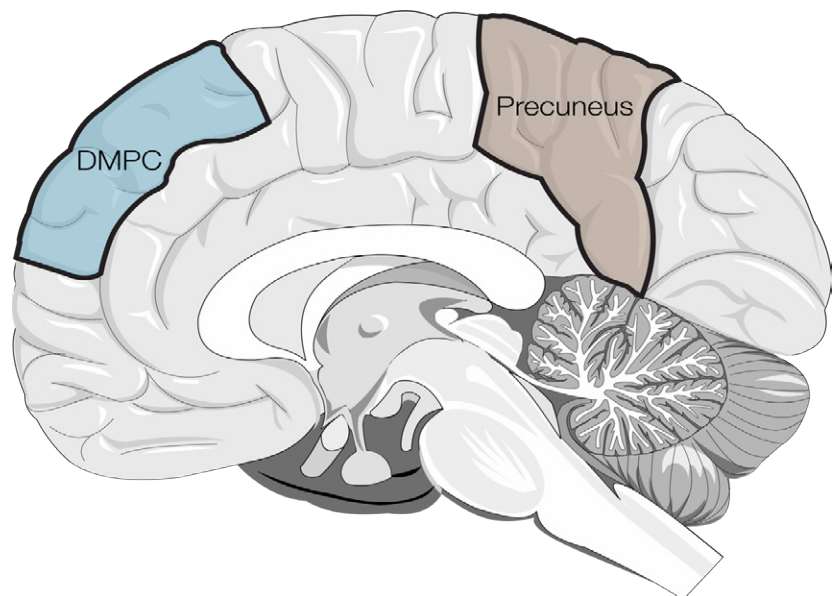


Figure 2: Activity in the dorsomedial prefrontal cortex (DMPC) and precuneus predicts success rate of tailored smoking cessation plans in smokers. The DMPC is responsible for higher-order cognitive functions such as planning, processing information, and many others.¹¹ The precuneus plays roles in retrieval of episodic memory, self-processing, and self-consciousness.¹²

brain activity. Four months later, the researchers were successful: greater activity in the dorsomedial prefrontal cortex and precuneus predicted smoking abstinence (Fig. 2).⁴ Further research regarding cognition may use similar techniques in order to identify parts of the brain that are more susceptible to good arguments.

ANTI-VACCINATION

One argument that has become prominent in today's mainstream discourse is that of vaccination. Although most approve of vaccination, a minority of people claim that vaccines are harmful and refuse to vaccinate. This refusal has had major repercussions. According to a recent report released by the World Health Organization, there have been over 40,000 cases of measles in Europe this year—a two-fold increase from last-year, and an eight-fold increase from 2016.⁵ This reemergence has been ascribed to an increase

in the number of parents who advocate for anti-vaccination, a belief that initially arose based on invalidated research that erroneously claimed that vaccines could cause autism.⁶ The drastic increase of late in vaccine-preventable diseases demonstrates that greater efforts are required to shift anti-vaccination attitudes. But how can scientists accomplish such a feat?

One reason that parents refuse vaccination is because they believe that vaccines have the potential to harm their children.⁷ Consequently, strategies geared toward countering this belief have focused on rationalizing the safety of vaccines by, for instance, providing scientific explanations of their ingredients. Unfortunately, these attempts are often futile, due in no small part to the psychological phenomenon of confirmation bias.⁸ In short, people who hold strong beliefs will often inflate the importance of evidence supporting their views and ignore evidence contrary to these views. Confirmation bias often drives how people

"According to a report released by the World Health Organization, there have been over 40,000 cases of measles in Europe this year—a two-fold increase from last-year, and an eight-fold increase from 2016."

perpetuate stereotypes, form opinions, and make decisions. Fortunately, scientists have tested other methods. Rather than trying to refute anti-vaccination claims, researchers have tried replacing those beliefs with new information about the health risks of not vaccinating. They found that out of several arguments, arguments that informed subjects of the risk of disease caused subjects to favor vaccination the most.⁹ The anti-vaccination mindset is not necessarily as set in stone as it may appear.

Further research, however, is still required to create effective strategies that can counter attitudes on anti-vaccination. A recent study of over 5,000 people discovered that anti-vaccination attitudes were high among those who exhibited individualistic or hierarchical worldviews among other beliefs, while other demographics such as education or income level had little correlation with anti-vaccination attitudes.¹⁰ Methods and arguments for correcting anti-vaccination attitudes could target some of these other belief systems. Research on anti-vaccination attitudes also contributes to efforts of understanding people who are unfazed by evidence-based refutations, such as climate-

change deniers or Flat Earth theorists. So although anti-vaccination is a mounting problem, society is not completely defenseless in countering its effects.

THE RHETORIC OF SCIENCE

The results of the above studies call into question certain approaches that members of the scientific community may follow in argumentation and discourse—in other words, the rhetoric of science. Scientists often laud science for its precise, unemotional rationality, and its conception as an objective and universal language by which people the world over can discuss observations of the surrounding world. When people express polarized political beliefs through primarily pathos-based arguments on Facebook, many people claim that their closed discourse induces the formation of echo chambers that are uncondusive to fair rhetoric. Yet, when academics communicate verbose ideas inaccessible to the general public using the language of logos, isn't a separate echo chamber formed, one that resonates with remarks floating through the halls of academic ivory towers? The type of conversation that

happens in Facebook groups and scientific journals may be distinctly different in their content, but they are not quite so in their form.

It should come as no surprise that the credibility and argumentation of scientists in public discourse is disputed. Scientists may take pride in their objectivity, but this language that is restricted to academics contributes to a rhetoric that strives to be primarily based on logic alone. Good argumentation employs a variety of rhetorical modes of persuasion. Evidently, replacing the fear of vaccines with a more intense fear of disease risk conveyed using scientific reliability changed attitudes more effectively than any solely science-based argument could. And in this era replete with crisis and uncertainty, scientists must remember that although science plays a key role in everyday discourse, it is merely one component among many anyone should use when communicating with the rest of the world.

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THE **CRISIS** AND CONVENIENCE OF SYNTHETIC PLASTICS

BY MINA NIKATANI

“I just want to say one word to you. Just one word... plastics. There’s a great future in plastics.” Back in 1967, a movie titled *The Graduate* expressed this rather insightful sentiment. This quote very nearly predicted the future; indeed, in the modern world, plastics are practically ubiquitous. Most marketplaces bag groceries in plastic bags. Families store leftover food in plastic food wraps and containers. Transparent plastics have even replaced the glass of some windows and most bottles. In many ways, overstating the importance of plastics is difficult. However, this material, which seemed so revolutionary fifty years ago, has caused many problems that the quote from *The Graduate* could not have foreseen. Today, a quick Internet search reveals heartbreaking photos of animals caught and killed by the plastic waste that people carelessly leave around. That same plastic has also created the infamous island of trash floating around somewhere in the Pacific Ocean, an obvious problem with a less than obvious answer. While synthetic plastics have certainly had some positive impact on our society, their alluring convenience caused a new kind of

crisis in waste production.

Calling this waste production a crisis is no understatement. Globally, the annual production of plastics exceeded 300 million tons in 2015. About 34 million tons of this plastic was turned to waste and over 90% of this waste (which is nearly one hundred times the weight of the Empire State Building) ended up in landfills or the ocean.¹ With the global population on the rise, the amount of generated plastic waste will likely continue to grow. With greater waste come greater problems—a crisis in the making, if not one already, due to the inherent toxicity associated with the chemistry of plastics.

Plastics are synthetic polymers—long chains of smaller molecules bonded into a macromolecule—often derived from petroleum. These polymers exhibit uniquely exploitable characteristics such as strength, flexibility, or a combination of both. The plastics used in packaging or consumer goods are special in part because they are considered safe and non-toxic.² Because consumer plastics are quite inert, people can use them for a variety of purposes and do not need to worry about, for example, their

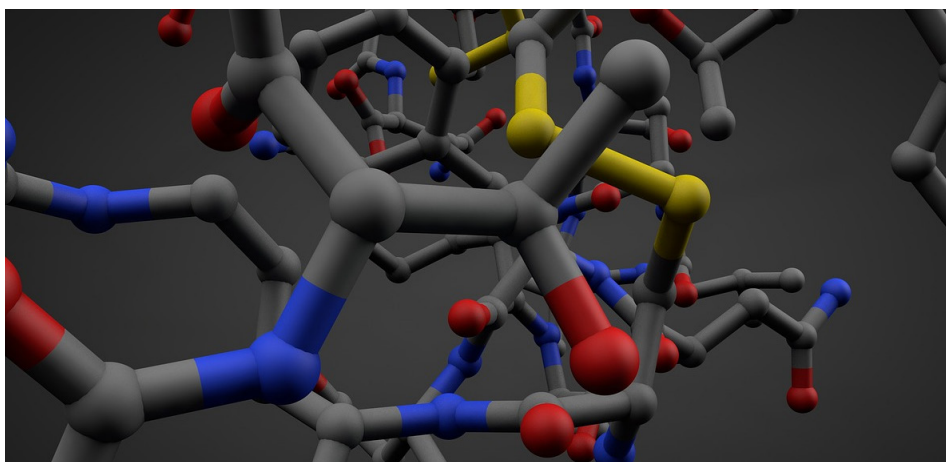


Figure 1: Polymer structure. Polymers are macromolecules—long chains of many atoms strung together to yield unique properties.

food wraps poisoning them. In this regard, most consumer plastics provide short-term convenience.

However, in the long term, nothing is perfect, and plastics are no exception. In fact, polymerization—the process of making plastics by stringing together monomers (individual small molecules) into a long chain—is rarely complete. Unreacted monomers and other additives, such as plasticisers or catalysts, sit in the material without being bonded to the polymeric chains.² These molecules are capable of seeping out of the material, especially when plastics are left in marine environments, which creates numerous public health hazards. For example, research reveals that vinyl chloride, the monomer of polyvinyl chloride (PVC), is carcinogenic, and benzyl butyl phthalate, a plasticiser used in PVC,

can cause reproductive harm.² PVC is the third-most widely produced plastic in the world.

The toxicity associated with plastics also causes negative ecological effects. Because plastics make up a large portion of marine waste, plastic debris are likely to degrade in water and break apart into smaller pieces. Most of this waste exists as microplastic—pieces of plastic smaller than 5 millimeters, which is about a quarter the size of a penny.³ Animals can easily ingest microplastic, which acts as a carrier for a variety of toxic chemicals and builds up in tissues. This build-up can cause pathological stress and impede reproduction.³

To solve the problem of toxicity associated with plastics, scientists are investigating biodegradable and bio-based polymers, which are considered environmentally-friendly. Biodegradable polymers break down into carbon dioxide and water, while bio-based polymers are made of renewable resources.⁴ At the forefront of this research is polylactic acid, a polymer that can be synthesized from only sugar and corn. Polylactic acid possesses a high transparency while remaining resistant to dissolving in water.⁵ To replace most conventional plastics, the newly developed environmentally-friendly materials will need to possess both of these characteristics—transparency and dissolution resistance.⁵ Other experiments have identified a natural polymer, poly- β -hydroxybutyrate (PHB), produced by bacteria.⁶ Bacterial cells effectively func-

tion like “mini-factories” that produce this completely biodegradable material.⁶ Both polylactic acid and PHB present apparently suitable replacements for conventional plastics.

Unfortunately, many biodegradable and bio-based polymers share faults that hamper their implementation by society. Perhaps the most pressing issue is the fact that these polymers do not display the same properties as their synthetic counterparts. Realistically, to be used effectively and achieve desirable properties, these environmentally-friendly polymers would require blending with conventional, synthetic polymers.⁴ But while the addition of synthetic polymers mitigates issues surrounding the use of conventional plastics, it does not eliminate them.⁴ Another issue is that environmentally-friendly polymers cannot yet be programmed to degrade at a specific time. For this reason, they cannot be used reliably for storage.

Nonetheless, polymer-decomposing microorganisms may provide an answer to the crisis of plastic waste. Researchers are considering natural enzymes as a solution to break down synthetic polymers.⁷⁻⁸ Although this idea is feasible, it suffers from several limitations. The decomposition process initiated by enzymes takes place only under specific conditions. Furthermore, enzymes can only break bonds in the portion of the chain that has a specific molecular composition and a specific arrangement in space.⁷ In other cases, enzymes act more readily on polymers that are already more liable to decompose in water.⁹ In short, decomposition of current conventional polymers is possible but still impractical.^{10,11}

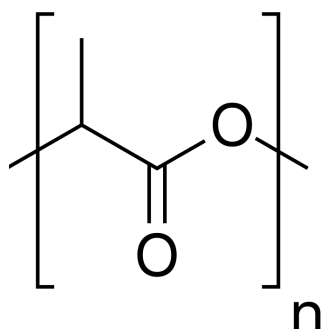


Figure 2: Chemical structure of polylactic acid (PLA). Synthesized from corn and sugars, PLA is a candidate for replacing synthetic plastics.

“While it is true that plastics have had a positive impact on the world, their convenience has caused a new kind of crisis in waste production.”

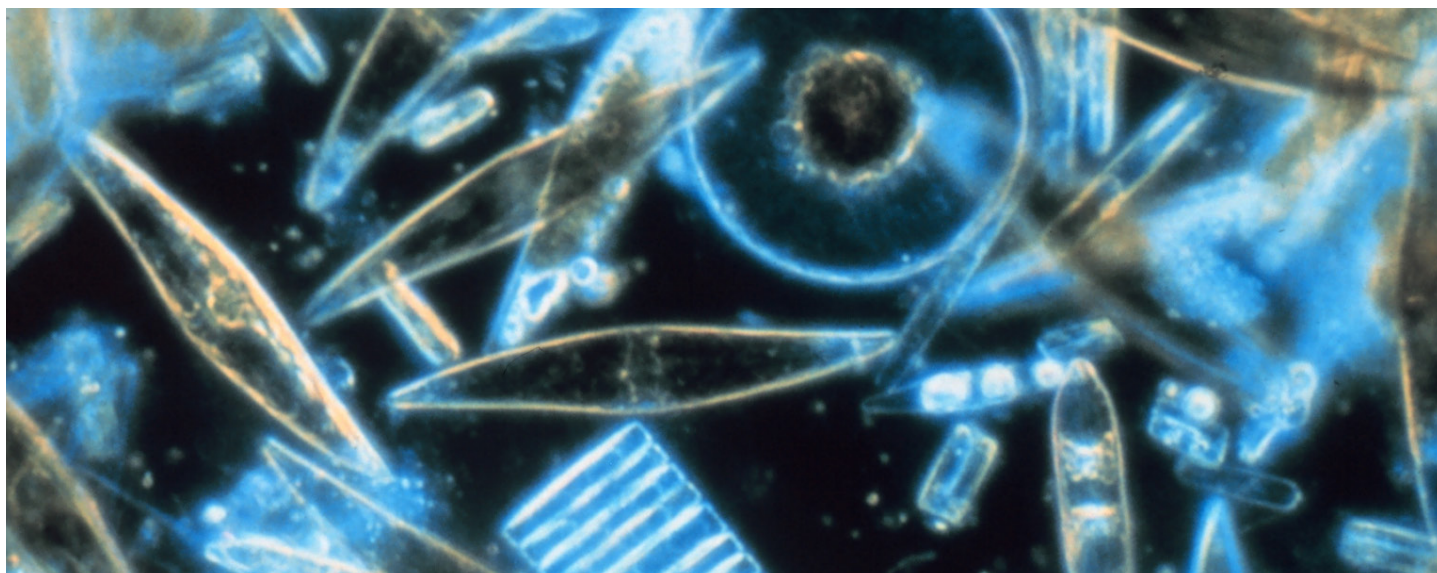


Figure 3: Microorganisms. Scientists are currently considering using microorganisms and enzymes to break down polymeric waste.

In many ways, plastics have lived up to their potential—to such an extent that solving their inherent problems introduces new ones. Until researchers find a solution with all of the benefits and none of the dangers, society will continue to face the dilemma of pitting plastic waste crisis against the comfort that comes with cheap convenience.

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"Addition of synthetic plastics still poses a problem because it does not fully resolve the issues surrounding the use of conventional plastics."



THE BELO MONTE DAM: GREATEST “NATURAL” DISASTER OF OUR GENERATION?

BY SHANE PUTHUPARAMBIL

In 1989, in the Brazilian town of Altamira, a Kayapo woman spoke passionately to a gathering that had been arranged by various international nonprofits. “We don’t need electricity; electricity won’t give us food,” she said. “We need the rivers to flow freely—our futures depend on them. We need our forests to hunt and gather in. Don’t talk to us about relieving our ‘poverty’—we are the richest people in Brazil. We are Indians.”¹

Strong-willed and emotional, the Kayapo woman’s voice reverberated throughout the international community.^{1,2} Protesting the Brazilian government’s plans for several hydroelectric projects on the Xingu River, the Kayapo (and other tribes) had forced the World Bank to scrap the loans for the dams and pushed back the building plans for

nearly two decades. However, in 2011, the Brazilian environmental ministry (IBAMA) granted licenses to Norte Energia—a Brazilian construction consortium—to start construction on a new project. Today, the world’s fourth largest hydroelectric project, known as the Belo Monte Dam, is nearly complete, and the social and environmental concerns of the past are now the nauseating realities of the present.

THE XINGU AND BELO MONTE

The Belo Monte hydroelectric project is positioned on the lower Xingu River, in a particularly fast-flowing region that is commonly referred to as the Volta Grande or “the Big Bend.”^{3,4} Areas where water is

shallow and traveling at high velocities are often referred to as “rapids.” The Volta Grande represents some of the largest and most complex rapids on Earth.⁵ Prior to human development, this bend was home to hundreds of freshwater fish species, each inhabiting its own unique niche within the river. In fact, a recent survey collected an astounding 450 species from 48 distinct fish families in the Volta Grande, demonstrating the enormous diversity of fish in the river.⁶ The Belo Monte hydroelectric complex, which is made up of two dams, was designed to harness the incredible rush of water by redirecting the Big Bend through a series of hydroelectric turbines.⁵ This ambitious project would come with costs: creating a 260-mile reservoir, submerging approximately 150 square miles of rainforest in water, harming aquatic ecosystems, and displacing about 30,000 people.⁷ In essence, the project destroyed the balance maintained for thousands of years between the indigenous people and their wilderness, resulting in the demise of a region once revered for its cultural and biological diversity.

“The project destroyed the balance maintained between the indigenous people and their river, resulting in the demise of a region once revered for its biological and cultural diversity.”

IMPACTS ON THE XINGU RIVER

The new dam will have countless negative effects on the overall biology of the river. The Belo Monte complex will ultimately harm the habitats of hundreds of endemic fish species, primarily affecting specialists, species that are intolerant to changing environmental

Because many of these fish live only in the Xingu River, projects like the dam can jeopardize the existence of entire species. Almost all species of rapid-dwelling fish have severely diminished in number, particularly due to the decrease in water flow and increase in surface temperatures. Numerous catfish of the species *Baryancistrus xanthellus* were found dead in the upstream section shortly

after the dam's reservoir was filled in 2016. Downstream of the Belo Monte complex, researchers found that the generalist species, fish that easily adapt to immense changes in the environmental conditions of a system, replaced non-tolerant specialist species.⁶ The extinction of specialist rapid-dwelling species, especially in a region where divergent evolution is highly active, will be an immense loss not only to science, but also to the fishermen and nearby cities that rely on the ornamental fish trade to drive the local economy.



Figure 1 (left): Rapids typical of the Volta Grande, characterized by the vegetation along the banks and the huge partially submerged boulders. Fishermen find plecós, medium-sized suckermouth catfish, wedged between crevices formed by the rocks and boulders.

Figure 2 (right): The main dam shortly before its completion.

conditions. The dam project will damage each part of the river on which the hydroelectric complex lies: the upper section, middle section, and lower section. Upstream, the dam has already slowed the rapids, and as a result, the substrate—the river bed—will continue to erode significantly. Other effects include an increase in surface temperatures and a lower dissolved oxygen content in the water. Downstream of the powerhouse, the overall water flow will continue to decrease, and the quality of the water will worsen due to the sedimentation, erosion, and increased temperatures of the system upstream.⁶ The construction of the Belo Monte dam raises several environmental concerns, as evidenced by a large decline in the populations of endemic fish.



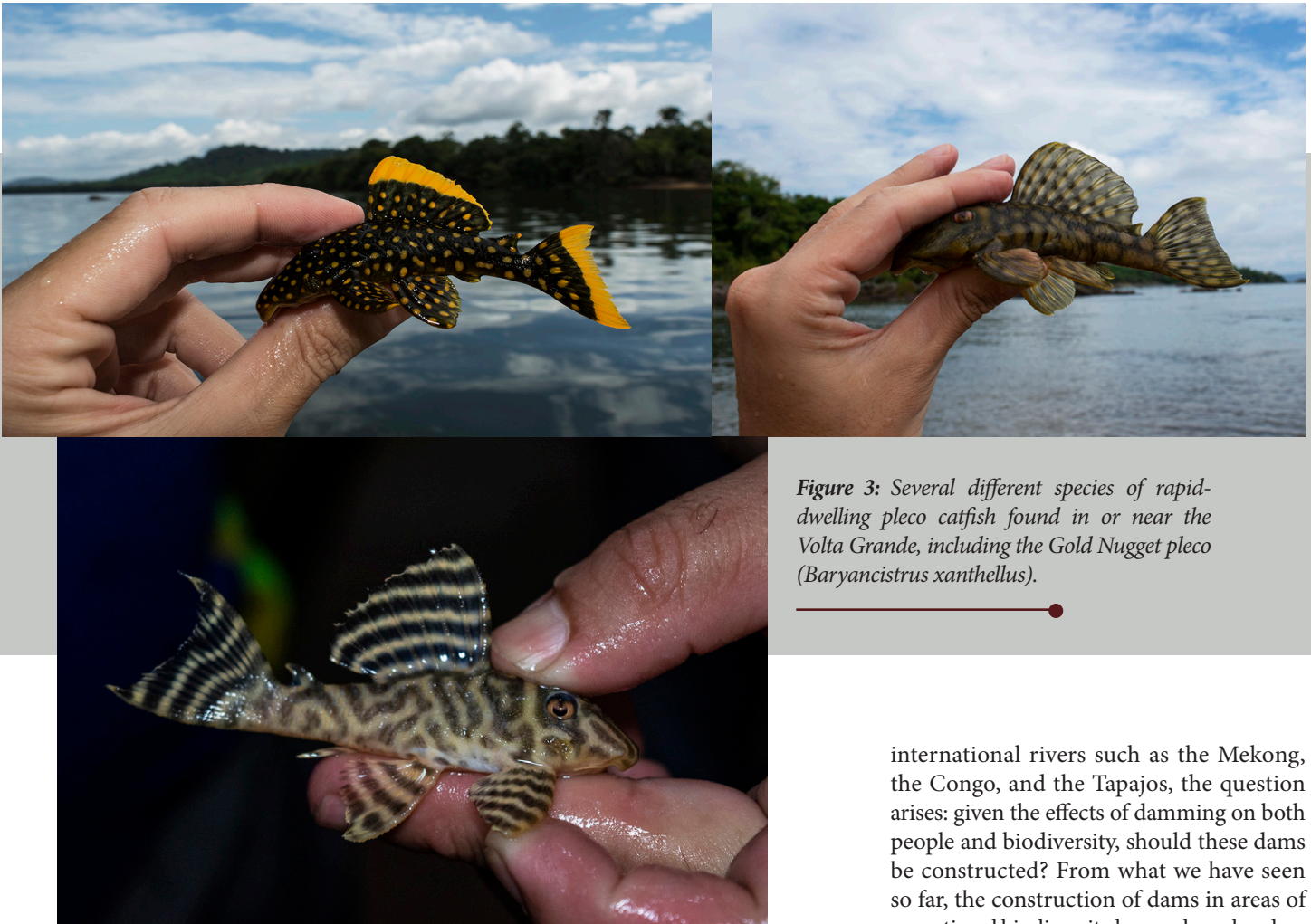
"The extinction of specialist rapid-dwelling species, especially in a region where divergent evolution is so active, would be of immense loss to not only science, but to the fishermen and nearby cities who rely on the ornamental fish

IMPACTS ON FISHERMEN

Besides the impacts on the ecological and geological structure of the Volta Grande, the Belo Monte hydroelectric complex will also cause great harm to the ornamental fishing industry. Ornamental fishermen collect the colorful fish from the Xingu River to export for the global pet trade, but after the

fish has notably declined. In addition, the decreased water levels exposed key fishing grounds along Volta Grande to overfishing. Hence, fishermen have to collect fish in deeper parts of the river where the current is stronger. Consequently, it often takes significantly more time to collect a sellable quantity.⁸ The ornamental fishery as a whole is becoming increasingly insufficient to support the families of the fishermen, so

and the full, long-term consequences of its construction will not be known for years to come. Scientists have been making extensive efforts to document the effects of the dam on the Xingu River, with the hope that this information can persuade governments to pursue sustainable means of energy generation and to avoid making the same mistakes again.⁵ With other megadams being planned for construction on other



*Figure 3: Several different species of rapid-dwelling pleco catfish found in or near the Volta Grande, including the Gold Nugget pleco (*Baryancistrus xanthellus*).*

construction of the dam, the number of fish trading companies dropped from 25 to 4. This illustrates the challenge that ornamental fishermen endured before eventually going bankrupt and leaving the industry. In 2014, Van Hall Larenstein University conducted a survey of the remaining fishermen. The researchers found that the abundance of the collectable fish and the overall health of these

much so that many decide to go and work in construction or cattle ranching elsewhere. This adds to the snowball effect often induced by environmental destruction, as activities such as ranching, farming, and construction often involve clearing forests and damaging other natural resources.

The Belo Monte Dam has cost the wildlife and people of the Xingu River considerably,

international rivers such as the Mekong, the Congo, and the Tapajos, the question arises: given the effects of damming on both people and biodiversity, should these dams be constructed? From what we have seen so far, the construction of dams in areas of exceptional biodiversity bears a large burden, yet the recent election of Brazilian president Jair Bolsonaro, a pro-dam advocate, has made the development of alternatives unlikely. That being said, it becomes even more clear that in order to preserve the Amazon, a new, truly sustainable and renewable energy source, is greatly needed.

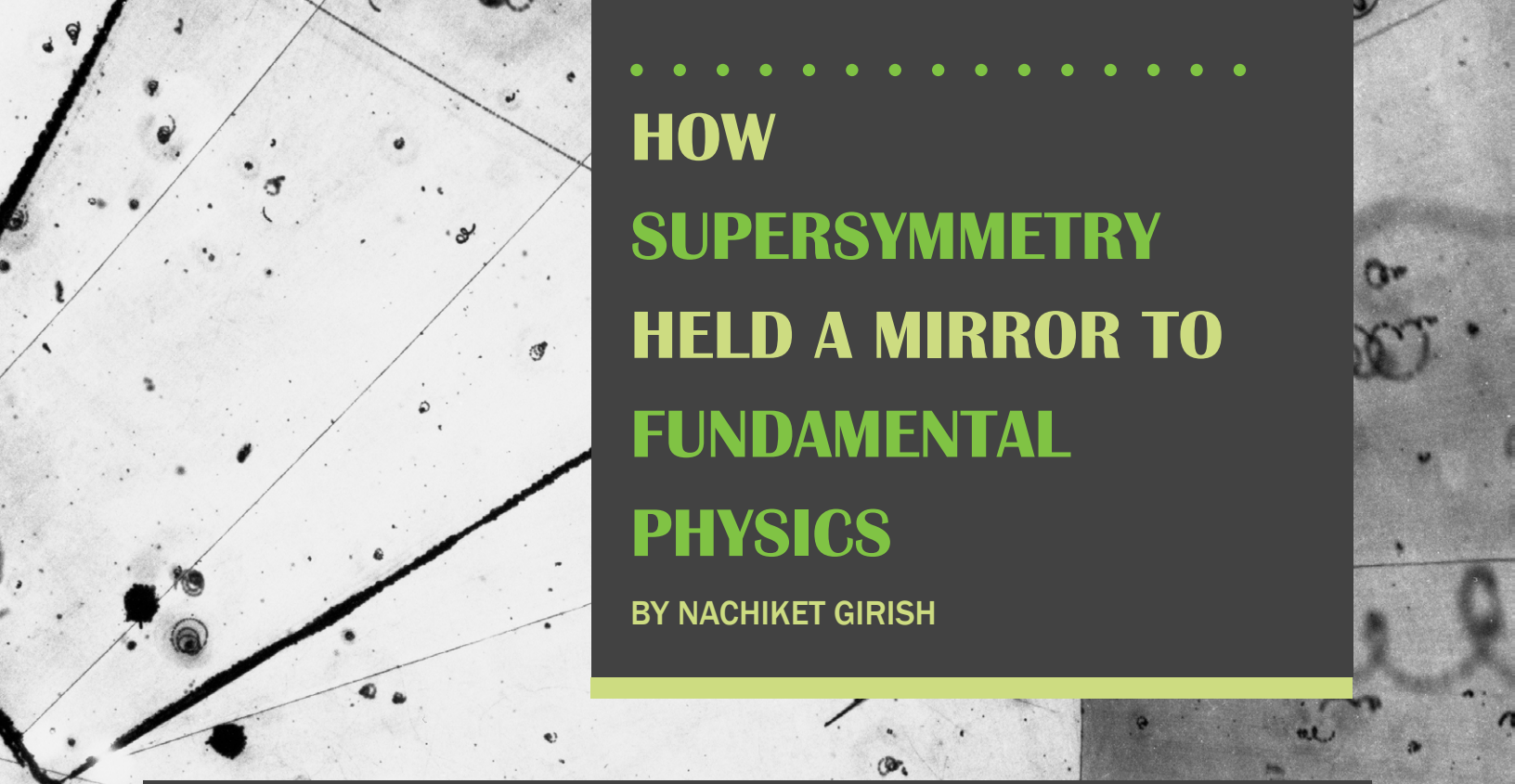


Figure 4: Daniel, a young fisher in training, holding some gold nugget plecos that he collected.

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HOW SUPERSYMMETRY HELD A MIRROR TO FUNDAMENTAL PHYSICS

BY NACHIKET GIRISH

THE CURRENT DEADLOCK OF SUPERSYMMETRY HAS RAISED NEW QUESTIONS
ABOUT WHAT MAKES A SOUND PHYSICAL THEORY

If you have seen physics in the news lately, you likely get the impression that now is an exciting time to be a physicist. Just a few months ago, the Nobel Prize for physics was awarded to a woman, Donna Strickland, for the first time in 55 years. The observation of gravitational waves three years ago heralded a whole new era of observational astrophysics. Three years before that was the massive triumph of particle physics with the discovery of the Higgs Boson.

What you might not remember, however, is what these discoveries represent. The observation of gravitational waves was the verification of a prediction Einstein made a hundred years ago, while the Higgs Boson was experimental confirmation of a fifty-year-old hypothesis. This symbiotic relationship between theory and experiment is the defining principle of science.¹ Theorists are the pioneers who plow through unexplored routes in search of new destinations, while experimentalists check every step to evaluate whether the theorists are heading in the right direction.

But even as we celebrate these monumental achievements of science, physics itself is going through a period of uncertainty, with one of the hottest theories of particle physics—

supersymmetry—increasingly finding no support from experimental data. The debate over how to explain the lack of support for supersymmetry has shaken the very foundations of scientific philosophy.

THE STANDARD MODEL AND BEYOND

Our story begins in the 1970s, with the development of the Standard Model, the broadest and most successful quantum theory physics has ever seen.² This theory explains almost every single phenomenon we can observe and has justifiably been called “the pinnacle of human achievement.”²

Despite the Standard Model’s great success, however, it is not bereft of problems. One of its most significant difficulties is known as the “hierarchy problem.” Theoretical calculations of the mass of the Higgs Boson and other related particles have revealed a troubling difficulty—quantum corrections should have caused the masses to be far, far greater than what had actually been observed.³ Quantum corrections are terms which theorists must add to their equations when solving problems using a method known as perturbation theory. In this method, a complicated problem is solved by first writing the solution for the

simplest case of the problem, and then adding further terms—the quantum corrections—to take into account the more complex features of the problem which the original solution had ignored.⁴

One possible but highly controversial explanation was that these corrections fortuitously canceled each other out. Alternatively, it was possible that there was a hidden mechanism which balanced them out. Attempts to solve this problem led to the development of one of the most exciting new tools of theoretical physics—supersymmetry.

Developed in the 1970s by several physicists, supersymmetry postulates that every matter particle or *fermion* has a corresponding partner which is a force particle or *boson*. A photon of light, for instance, is a boson, while an electron is a fermion. These partner particles are thought to cancel the quantum corrections caused by the regular particles.⁵ Not only does supersymmetry thus neatly resolve the hierarchy problem, it has the added benefit of proposing an explanation for dark matter—the mysterious class of matter we know exists but have yet to observe—by offering the supersymmetric partners of our regular particles as possible dark matter candidates.⁶ It almost seems too good to *not*

Standard Model of Elementary Particles

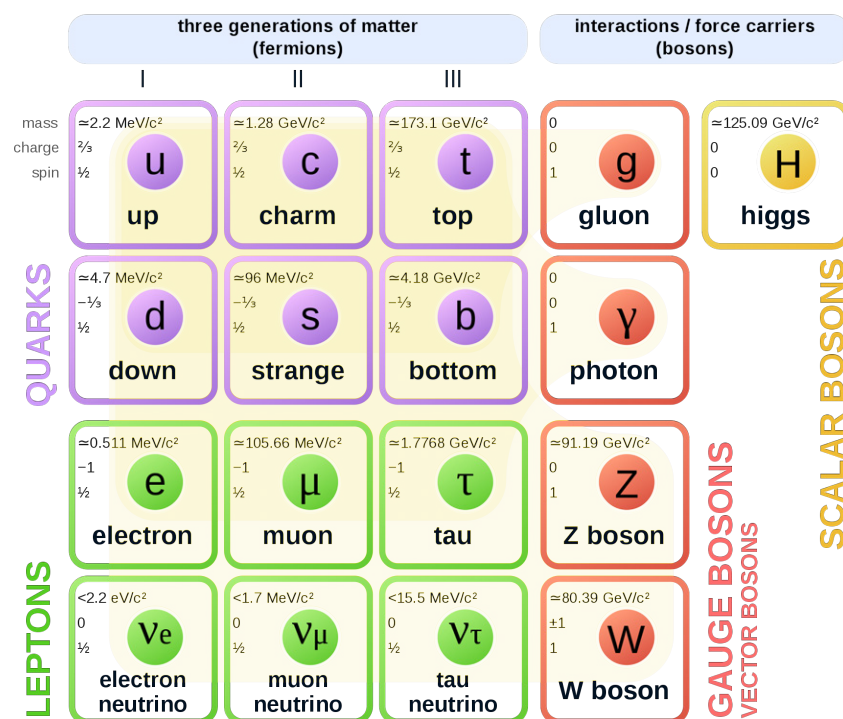


Figure 1: The standard model, the theory of almost everything.

be true.

The problem was that verifying the predictions of supersymmetry would require particle accelerators capable of reaching energies no accelerator of that time could reach. To remedy this, physicists built the Large Hadron Collider (LHC) in 2008. It was, and remains, the largest machine ever built, designed to generate sufficiently high energies to explore the new physics beyond the standard model.⁷

OPTIMISM TURNS TO DESPAIR

It is here that the mood of this narrative becomes less upbeat. In the ten years that the LHC has been operational, it has not detected a single supersymmetric or dark matter particle. Nor has it given any clues whatsoever for the existence of supersymmetry. The discovery of the Higgs, though a phenomenal success, was but an additional confirmation of the Standard Model.

The current state of affairs, affectionately dubbed the “nightmare scenario,” leaves physicists in a unique quandary. Supersymmetry has not been disproved—in fact, that outcome would have been

much more helpful, as it would have at least provided theorists with some direction. On the contrary, the experimental data neither supports nor disproves any of the predictions of supersymmetry. Though the LHC has been able to confirm and reaffirm the Standard Model, it has failed to fulfill its founding purpose.⁸ The upshot is that a large number of physicists are left with a theory they spent several decades developing to such a degree that even its critics acknowledge its mathematical potential—without any idea of its validity.

Supporters of supersymmetry suggest that physicists simply underestimated the masses of the supersymmetric particles; perhaps, the superparticles are actually heavier than what even the LHC can currently detect.⁹ Not only might some consider this a suspiciously ad hoc claim, but increasing the predicted masses of the superparticles also raises questions regarding the principle of naturalness.

NATURALNESS IN PHYSICAL THEORIES

Naturalness is a principle in scientific philosophy which demands that the physical parameters of a scientific theory should arise

as “naturally” from fundamental principles as possible; that is, there should be an underlying principle which explains all the predictions of the theory. According to this concept, such a theory is preferable to one in which theorists can artificially fine-tune the values of their predicted constants so that the constants conveniently combine to give the expected result.⁸

In the case of supersymmetry, in order to correctly cancel the quantum corrections to the Higgs mass, theorists required the superparticles to be more or less equal in mass to their partners. But as the LHC kept reaching progressively higher energy and mass scales without finding any superparticles, hopeful supersymmetry proponents claimed even higher masses for the partner particles. In order to salvage the situation, supporters of supersymmetry had to fine-tune the values of a variety of other constants, so that when combined, they still yielded the correct Higgs mass.³ Paradoxically, at some point this explanation amounts to the same reasoning inherent in the non-supersymmetric Standard Model—that the serendipitous cancellation of the quantum corrections, or in this case, the superparticles attaining the required fine-tuned values, is just a coincidence.

While in principle there is nothing wrong with this kind of theory—we might simply be lucky to live in a universe where fundamental constants are custom-made for our existence—it flies in the face of the principle of naturalness, which demands an explanation for this coincidence. In following this approach, moreover, proponents of supersymmetry can keep raising the target particle-mass level for experimentalists to

“Even as we celebrate these achievements of science, physics itself is stuck in a period of uncertainty, with one of the hottest theories of particle physics—supersymmetry—finding no support from experimental data.”

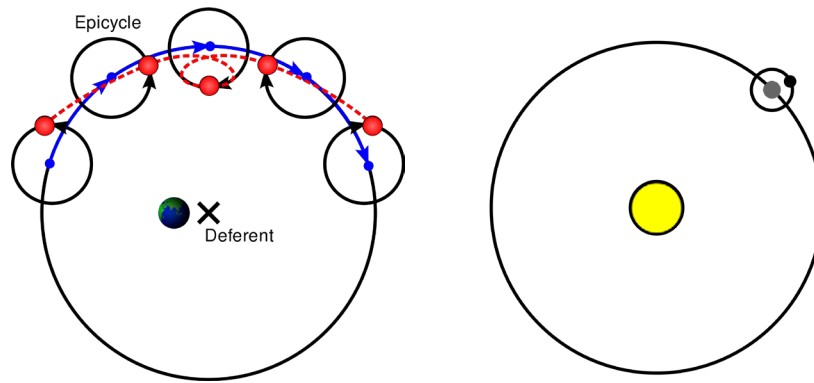


Figure 2: Ptolemy vs. Copernicus. Since a large number of epicycles of very convenient sizes and orientations need to be added to get the required accuracy in Ptolemy's model, we consider his geocentric theory to be fine-tuned and "unnatural." We thus prefer the much simpler heliocentric theory proposed by Copernicus.

reach by simply fiddling with their constants—and thus keep justifying their research (and their funding). These modifications make the theory more fine-tuned and thus less natural and elegant.

The trouble is that the concept of naturalness is purely human, and does not necessarily have anything to do with the universe we live in. Indeed, when there is no experimental evidence clearly falsifying one theory or another, the choice of which one to use often becomes a matter of taste. For instance, in discussing a possible modification to supersymmetry in his paper "The State of Supersymmetry after Run I of the LHC," author Nathaniel Craig, a physicist at the Institute for Advanced Study at Princeton University, evaluates the modified theory by noting that "there is no reason it can't be there, but it's fairly unsatisfying as a theory

of nature."⁸ There is, however, no quantitative reason to reject a theory simply for being "unsatisfying."

On the other hand, historically, similar cases have demonstrated that the simpler theory is usually correct. For example, Ptolemy clung to the geocentric idea of the solar system by proposing that heavenly bodies moved around the earth in multiple nested circles called epicycles. By arbitrarily adjusting the number and size of epicycles in his model, he could make his theory as accurate as desired.¹⁰ In contrast, Copernicus proposed a much simpler heliocentric model of the solar system, with Newton later providing the underlying physical explanation for this model. Despite the matching accuracy of both theories, Copernicus' more elegant heliocentric model of the solar system is what ultimately proved to be the more correct one. Be it theories of

the solar system or those of the expanding universe, scientists have had to develop new theories whenever the reigning ones have required overly contrived modifications.¹¹

This dilemma leaves physics at an interesting crossroad, one in which the debate over the validity of a theory is, to a certain extent, philosophical rather than scientific in nature. Young researchers entering the field now have a difficult choice to make: continue efforts at justifying the absence of experimental evidence by fine-tuning—hoping that some evidence shows up sooner rather than later—or break away and explore radically different ideas. Which way should future research go? This debate, regardless of how it is resolved, will have an immense impact on the future of theoretical physics. It is indeed an exciting time to be a physicist.

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CLIMATE CHANGE AND THE NUCLEAR OPTION

BY MATT LUNDY

Unlike the theoretical dangers of nuclear holocaust or worldwide pandemic, climate change is a real threat that might soon cause irreversible devastation to humanity. Climate change is happening now, and if we wait for its more overt effects to be revealed before we act, it might be too late to avoid disaster. As such, it is of vital importance that policymakers and the general public alike understand the urgent threat it poses and how best to tackle it.

There is an overwhelming amount of evidence to suggest that current climate change is being caused by human activity. For roughly the last 150 years, the Earth has been rapidly getting hotter (Fig.1).¹ This temperature increase lines up with the huge amounts of carbon dioxide that fossil fuel consumption—as a byproduct of burning coal, oil, and natural gas—and deforestation—resulting from decreased carbon absorption—have released into the atmosphere over the same time period. As a greenhouse gas, carbon dioxide in the atmosphere absorbs and re-emits infrared

radiation, which causes warming. With no other likely candidate as a plausible cause of the huge increase in temperature (Earth's orbit, the sun, volcanoes, ozone and aerosol pollution all fail to fit the bill), human-produced carbon dioxide has taken the mantle of responsibility for the recent global warming.²

Because the idea of anthropogenic climate change is substantiated with such strong evidence, it comes as no surprise that the scientific community is almost unanimously in agreement regarding the theory's validity. In a meta consensus study spanning six independent studies, Assistant Professor John Cook at the Center for Climate Change Communication at George Mason University, along with over a dozen others, confirmed that 90-100% of publishing climate scientists agreed that humans were responsible for recent global warming.³ These results lend strong support to the oft-cited statistic that 97% of climate scientists agree with anthropogenic global warming (AGW). Another study con-

firmed that the side critical of AGW makes up a “vanishingly small proportion of the published research.”⁴

The effects of AGW are frightening to say the least. In the worst case scenario, where we take no action at all, temperatures would rise at the same rate they have been rising at thus far. A temperature change of just two to five degrees is enough to drastically heat up the planet; since 1880, the global average temperature has already risen roughly 0.8 degrees Celsius.⁵ Although a two to five degree change may seem minimal, the amount of heat necessary to achieve that average temperature difference across all of the land, oceans, and atmosphere of the Earth is monumental. Indeed, humanity is already seeing the effects of this temperature rise—from smaller ice caps and rising sea levels all the way to ocean acidification.⁶ These effects will worsen with more heat: rising sea levels will begin consuming coastlines and pushing people inland, while ocean acidification will destroy reefs and have a devastating impact on underwater

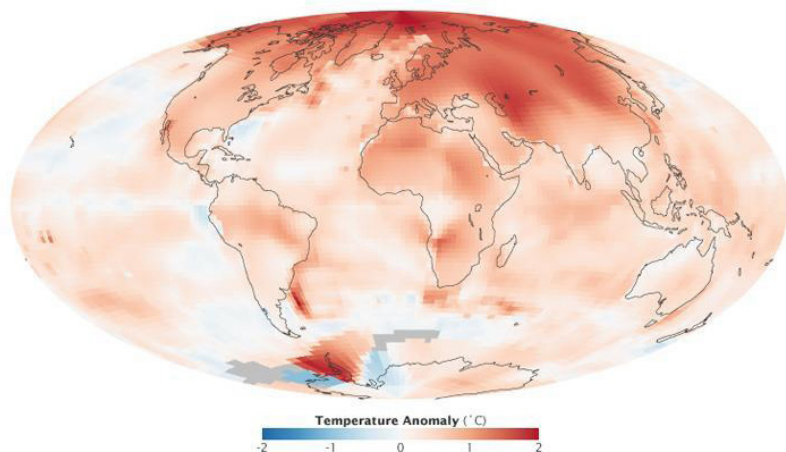


Figure 1: NASA representation of how current temperatures around the world compare to the average temperature since the late 1800s.²⁰

food chains.⁷

Accepting the existence of climate change allows us to explore avenues to combat it. While traditional renewable energy sources such as solar, wind, hydroelectric, or geothermal power are standard solutions, they each have their own drawbacks. Solar and wind energy require immense battery stores to be viable primary contributors to a large power grid. Meanwhile, hydroelectric energy, which arises from the natural movements of water, and geothermal energy, which originates from inside the Earth, are location- and resource-specific. Despite these challenges, these energy sources offer ample power and are undeniably cleaner in terms of carbon dioxide emissions than either coal, oil, or natural gas. In fact, many countries power themselves with these renewable energy sources, like Iceland and its use of geothermal energy, British Columbia and hydropower, Uruguay and wind, and

Germany and solar.^{8,9,10,11}

However, one renewable energy source that often gets overlooked is nuclear power. Nuclear energy is generated from either splitting the nucleus of an atom or from fusing multiple nuclei together. The former process, known as fission, is how energy is generated in modern day nuclear plants.

People are often hesitant about nuclear energy due to its association with catastrophes—such as the atomic bombings in Japan and the meltdowns of Chernobyl, Three Mile Island, and Fukushima. While these events provide reason to reflect on how to properly and safely utilize nuclear power, the negative stigma that they have bestowed on what is in fact a remarkably clean source of energy is unfortunate. The greatest testament to nuclear safety is that it has the lowest deaths per watt hours of energy generated; nuclear energy causes far fewer deaths globally per Petawatt hour (90) than coal (100,000), oil (36,000), hydro (1,400), wind (150), or even solar (440).^{12,13}

It is worth noting that one of the greatest concerns regarding nuclear energy, namely the threat of meltdown as seen at Fukushima and Chernobyl, is almost entirely preventable. These large-scale failures were largely due to human error, resulting from key safety procedures and requirements being neglected.¹⁴ Overall, nuclear energy has a very low death rate of 0.1 per Petawatt

hour of energy. This means that even if the entire U.S. were powered by nuclear energy, there would only be around one death every other year due to energy generation. In comparison, there are roughly 10,000 deaths per year in the U.S. from coal alone.¹³ Nuclear energy, like anything, will never be completely foolproof, but with strong and well-enforced regulation, its drawbacks can be mitigated immensely.

In addition to its safety, nuclear power is also highly adaptable. It can be implemented anywhere that has enough space to build a power plant, and—much like a coal power plant—provides a steady stream of energy. Because of this, it circumvents the problems that afflict the other forms of green energy, such as the need for better battery supplies to make up for the volatility of wind and solar, or the geographic limitations to hydro and geothermal.

Even though nuclear energy may very well be the golden ticket out of climate change, many countries are hesitant to adopt it. A recent study revealed that although many Australians do see nuclear as a clean alternative, they are fearful of the possibility of a nuclear meltdown.¹⁵ In a global survey, 62% of participants opposed nuclear power to some degree. Even in France, where nearly all electricity is generated by nuclear energy (Fig. 2), 67% opposed this energy source in the aftermath of the Fukushima disaster.¹⁶ The sentiment against nuclear energy is strong in the U.S., too. Just recently, Californians voted unanimously to close down their last nuclear power plant, the Diablo Canyon Power Plant.¹⁷ Much of the stigma surrounding nuclear seems to be bred out of ignorance, which is understandable, as most people only hear about nuclear power when disaster strikes. This phenomenon often leads to a negatively biased view of nuclear power, making it more likely that people will oppose it. Indeed, a study on American public perceptions of nuclear power found that with greater education and understanding of energy issues, people were more likely to support nuclear energy.¹⁸

Currently, the public's perception of nuclear energy is founded on a lack of information. The threat of disaster, biased media portrayal, and an overall lack of understanding

"Even though nuclear energy could be the golden ticket out of climate change, many countries are hesitant to adopt it."



Figure 2: A nuclear power plant in Cattenom, France. France's nuclear energy accounts for 76.3% of its total electricity production. The stacks rising out of the four large towers in this image are made of steam and are harmless.

when it comes to nuclear has scared the public and policymakers away from a potentially planet-saving energy source.¹⁹ But the merits that nuclear has over its alternatives, coupled with the pressing threat of climate change, make it more than worthwhile to reconsider our attitudes towards nuclear energy.

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MOMENTS OF MANIA: EMOTION- RELATED IMPULSIVITY AND BIPOLAR DISORDER

*Interview with Professor
Sheri Johnson*

BY SHEVYA AWASTHI, MATTHEW
COLBERT, DOYEL DAS, MELANIE
RUSSO, KAELA SEIERSEN, AND
ELENA SLOBODYANYUK



Professor Sheri Johnson.



► **Figure 1:** One of Vincent van Gogh's most famous paintings, *The Starry Night*. Psychiatrists Hemphill and Blumer suggested that van Gogh had bipolar disorder.^{1,2}

Sheri Johnson is a Professor of Psychology and Director of the Cal Mania (CALM) Program at the University of California, Berkeley. She is also an affiliated faculty member at UCSF's Depression Center. Professor Johnson's research centers on understanding the triggers of mania and depression within bipolar disorder, and on emotion-related impulsivity more generally. In this interview, we discuss the study of emotion-related impulsivity in individuals with bipolar disorder as well as the development of new treatments for bipolar disorder.



their usual self. The human challenge of living a life like that was fascinating to me. A couple months into my internship, one of my clients said, “I don’t get it. You’re doing all this work on depression and we don’t even know what triggers episodes of BD. Isn’t that the kind of question you want as a scientist?” And for the next 20 years that became my driving question: why do people move into these episodes of mania?

BSJ: How heritable is BD?

SJ: BD is among the most heritable of disorders. The estimates of heritability of mania are 0.85 and above. These estimates are based on careful community-based twin studies, so there isn’t a biased sample. There’s a huge genetic component. One thing I want to say very clearly is that just because something’s heritable, that doesn’t mean you’re stuck forever in the same state. It’s still treatable, and we can still make a difference.

BSJ: Several of your studies focus on adolescents with BD. What is the significance of studying this age cohort?

SJ: BD often seems to come on during adolescence. Over time, those episodes have a pretty important role in somebody’s life. They can interfere with relationships and work. If you catch people early on, you may get closer to understanding the risk factors as opposed to their aftermath. Throughout our studies, we take a two-pronged approach. One is to conduct longitudinal studies on people who have already been diagnosed. We wait until they are well, compare their psychological traits to those unaffected, and then see how those traits predict the onset of episodes. This can get tangled up because the individual is already going through the disorder. Our other approach is to take people who haven’t had a full-blown episode but have signals that they are at risk. If you follow those people over the next 10 years, a group of them is likely to develop BD. When we see a psychological correlate in the at-risk group, we feel like the trait is not just entangled with the aftermath of a serious episode, and so we feel more confident that we are onto something.

BSJ: In one of your studies you investigated the connection between sleep disruption and impulsivity.³ Why did you decide to focus on this relationship?

SJ: Sleep helps your prefrontal cortex function better, it helps you self-regulate, and it helps you control your emotions. It has been shown that if you deprive somebody of sleep, they are going to be more impulsive. Impulsivity and sleep problems are two of the key features you see in BD. They have always been treated as two separate features, but we thought maybe they are not so separate.

BSJ: How did you get involved in the field of psychology and specifically in studying bipolar disorder (BD)?

SJ: I began college as a music major, and that was really demanding with low potential for full-time employment. I thought psychology classes were interesting and provided a more secure career path. I didn’t get interested in BD until the last year of graduate school. I thought I was going to be a depression researcher, and I wanted to do a one-year internship at Brown University because their program focuses on mood disorders. When I got there, I was assigned to work with people who had BD, and I was hooked. Imagine a group of people who are a little bit more energetic and sparkly than the typical population. Within a matter of hours, they can go through an episode that renders them startlingly different from

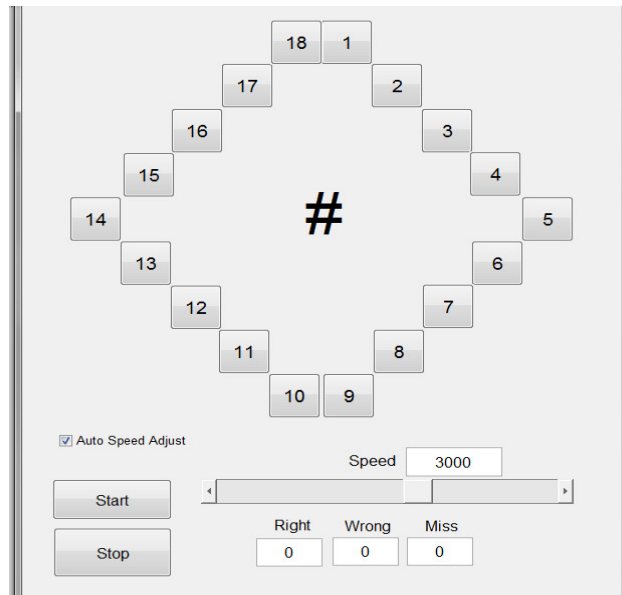


Figure 2: Example of the Paced Auditory Serial Addition Test (PASAT), a working memory task. Participants listen to numbers presented every 3 seconds one at a time and must add each number to the previous number. The participant's answer is entered by clicking the corresponding number on the screen. Image courtesy of Greg Siegle.

BSJ: In this study, you found that sleep disruption was associated with greater impulsivity in the BD group but not the control group. How can you explain this result?

SJ: One of the things with BD is that there is a greater vulnerability to certain kinds of challenges. If you have something that will change mood state, for most of us it might affect us only a little bit, but for individuals with BD it is likely to have more influence.

BSJ: A lot of your work centers on investigating emotion-related impulsivity (ERI). Can you define ERI and explain its importance in the context of psychopathology?

SJ: More than a decade ago, Whiteside and Lynam conducted a massive study to measure the different dimensions of impulsivity.⁴ There was a unique group of people who only became impulsive during states of high emotion, and during those states they would do and say things that they later really regretted. Whiteside and Lynam initially studied negative emotion, but of their protégés, Melissa Cyders, created a scale characterizing response to positive emotion states. It turns out that the negative and positive emotion scales are highly correlated; people with one tendency are very likely to have the other tendency. Furthermore, compared to any other self-rated form of impulsivity, ERI is much more predictive of psychopathology. The effects of ERI are stronger for conditions like aggression, depression, and anxiety. In our own work, we've

shown that ERI is very present for people with BD even after they remit. So it seems like ERI gives us much more predictive power in understanding psychopathology.

BSJ: In one study, you investigated neurocognitive mechanisms underlying ERI.⁵ What is response inhibition and how does it relate to ERI?

SJ: The basic idea of response inhibition is that you have to withhold the prepotent response. Why does that matter for emotion? Let's say you are driving down the road and somebody cuts you off. The impulse to go, "Ugh!" is prepotent. Not doing that, say if you were an Uber driver (who has to stay polite and serene), takes inhibiting your response. We know a lot about the neural circuitry involved in response inhibition, and it turns out to be the very same neural circuitry used when engaging in different forms of emotion regulation. It made sense to us to think that response inhibition could be one of the things going wrong for people with ERI. We and many other researchers have now shown that people with high levels of ERI show deficits in response inhibition.

BSJ: Why did you choose to focus on the effects of emotional arousal on response inhibition?

SJ: Most previous studies were just putting people into a standard response inhibition task. But where is emotion in that

“Compared to any other self-rated form of impulsivity, ERI is much more predictive of psychopathology.”

story? Clinically, people with ERI say that they are falling apart and can't constrain their behavior during states of high emotion. What do good and bad emotions—being euphoric and angry—have in common? High arousal. That was the next direction we went—looking at arousal and its influence on response inhibition for people with ERI. In two studies we did this through a mood induction. We brought people into the lab and showed them a scary or upbeat movie and then tested response inhibition. We could see that people with high levels of ERI were having problems with their response inhibition.

BSJ: Upon inducing a heightened emotional state in your participants, you measured arousal by pupil dilation. Why did you choose this over other measures such as heart rate?

SJ: One reason is that we frankly haven't had great effects with measuring heart rate. The other reason is that the pupil is pretty interesting. We know a lot about the circuitry guiding pupil dilation, and if you keep light levels constant, pupil dilation is directly influenced by noradrenergic systems in the brain (parts of the sympathetic nervous system that are highly involved in arousal), which have been really nicely traced out. Now you have this beautifully observable proxy for some of the noradrenergic spikes that might be happening in the brain.

BSJ: Finally, we read about your study evaluating a cognitive control training program for reducing ERI.⁶ Could you explain the cognitive training procedure, particularly the PASAT and Go/No-Go tasks?

SJ: Broadly, we've been trying to understand whether we can make a difference for ERI. There's a huge amount of literature out there that suggests that we might be able to strengthen some of these cognitive processes by practicing them. Andrew Peckham was the leader of this study, and he simply conducted six sessions of 15 minutes of practice of the Go/No-Go task and 15 minutes of the PASAT. The Go/No-Go task is a classic measure of response inhibition. We say to you, “Every time you see the letter X, we want you to press a button as quickly as you can. If you do that successfully, you are going to earn money.” We have you do that a lot of times so it becomes automatic. You see the X, boom!

You get very good at pressing the button. Then we say, “If you see a Y, don't press the button.” The PASAT is a working memory task where you have to attend to a sequence of stimuli that show up on the screen. It's very challenging, and people rate it as more unpleasant than a lumbar punch. That makes it seem like a terrible task to choose! On the other hand, if you wanted to look at your ability to use your prefrontal cortex during moments of distress, that's a perfect training task. You're catching people in a moment of frustration and asking them to get better at the skill. Andrew combined both of these tasks and saw that at the end of training, there was a lower score on ERI. He didn't see the same thing in the control group, and now he has five years to look more carefully at this training in a partial hospital program. We'll see how we do!

BSJ: Do you see these trainings becoming accessible to individuals beyond the study?

SJ: That's the great hope of cognitive remediation. One of the big controversies in the field right now is that when people were given access to a remediation program over the internet, it didn't seem to help. What happens there? Is it that when we shut down every other piece of stimulation in your life (like we can when you visit our lab), you rehearse more deeply? Is it that you work harder with the experimenter coming back to check how



Figure 3: Participant completing the Go/No-Go task. The participant is instructed to press a button in response to a symbol (“Go”) and withhold responses to another symbol (“No-Go”), which appears much more frequently. The task is a measure of response inhibition. Photo courtesy of CALM Lab.

you did? Or maybe when you push it out that broadly, you might not get the people you want to get? There's a lot of questions about what we need to do to disseminate. We've set up a web-based intervention on hyperarousal, but we're still keeping close tabs on it because we want to do a lot of interviews to evaluate how it works. Our hope is that it would be something very accessible for people to take from home, whether or not they feel like seeing a therapist.

BSJ: What are some limitations of self-report measures in your studies?

SJ: One limitation is that a lot of the items on ERI measures are about feeling regret about your activities. There may be people who show extremes of ERI but don't particularly regret it. There also may be people who are overly critical of themselves. They do something that's pretty modest, and they feel a deep sense of regret about it, even if it's exactly what most of us do during states of high emotion. One way researchers have gotten around that is to develop interviews from a parent's or friend's perspective, or to ask for observer ratings. The findings from those scales look very similar to the findings from the self-ratings. That's a good sign because it suggests that for the most part, people are pretty good informants about themselves on this front.

BSJ: How do different trainings for BD such as medication, cognitive training, and psychosocial training compare to each other?

SJ: At this point, every major treatment guideline worldwide would put medication first. We don't have anything that tops the effects of medication, so it's the first line of defense. Lithium is one of the first well-documented treatments for BD, and it's shown to be related to lower rates of suicidality. Most people are looking at the role of psychotherapies as adjuncts to medications. There's a lot of evidence that adding psychotherapy should be standard care—it helps people have a better quality of life, lowers the risk of relapse and hospitalization, and helps build back social and occupational roles. Unfortunately, psychotherapy is not always provided. We only have a couple studies of cognitive remediation for BD, and while there are some fascinating findings, I think we are in the earlier phases of understanding its effects for BD.

BSJ: What are the implications of your research in developing treatments for BD?

SJ: Much of my work for 20 years was focused on understanding reward systems in BD, and we have developed a pilot treatment that focuses on those facets. We need to test it more thoroughly, but we think that people can have a better sense of control by understanding that they may need to implement more emotion regulation strategies in situations with high rewards. We have not yet tested our treatments related to ERI and BD, but that's one of the next things we would love to do. We know that

“Our hope is that [cognitive remediation] would be something very accessible for people to take from home.”

this form of impulsivity is really important in BD; it predicts relapse, aggression, problems in functioning, lower quality of life, and greater risk of suicidality. If we could target ERI effectively, we might make a difference in some really important outcomes. I'm hoping that our work on ERI could be rapidly applied to BD and tested as an intervention.

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BRIDGING THE GAP BETWEEN THE FOSSIL RECORD AND THE MODERN DAY

Interview with Professor Seth Finnegan

BY CASSIDY HARDIN, AKASH KULGOD, MICHELLE LEE,
STUTI RAIZADA, AND NIKHIL CHARI

Seth Finnegan is an Associate Professor in the Department of Integrative Biology at UC Berkeley and a curator at the University of California Museum of Paleontology. He studies marine paleobiology and processes which shape marine ecosystems over time. We asked Dr. Finnegan about his favorite mass extinction event at the end of the Ordovician period (488 to 443 million years ago) and about the relationships that we can draw between past extinction patterns and the current anthropogenically caused extinction.



Professor Seth Finnegan.

BSJ: What interests you specifically in the late Ordovician period?

SF: The Ordovician is a very interesting period of time for a number of reasons. Most of the major animal groups that still exist today—mollusks, arthropods, etc.—make their first appearance during the Cambrian explosion. However, the majority of these groups are not particularly diverse. In the Ordovician period, which follows the Cambrian explosion, a number of these groups begin to diversify. By the end of the Ordovician period, most of the groups that ecologically dominate marine ecosystems for the next two hundred million years are in place. Then many members of these groups go extinct in a very unusual, rapid mass extinction event. Ultimately, the Ordovician an interesting period because it contains both a major diversification and a mass extinction event.

BSJ: What are rhynchonelliform brachiopods, and why did you choose to focus on them in your research?

SF: Fig. 1 shows some Ordovician fossils from our collections at the UC Berkeley Museum of Paleontology. Fig. 2 is a slab of fossils from my own field work in Quebec. All of these little fingernail-shaped things are rhynchonelliform brachiopods. In the Paleozoic period that ended with the Permian-Triassic mass extinction 250 million years ago, they were extremely common and diverse parts of marine ecosystems and the marine fossil record. If you want to study geographic patterns in the fossil record, you need a group with very high preservation potential, meaning every individual has a relatively high likelihood of ending up as a fossil. A lot of the groups we first think of when we hear the word “fossil” have relatively low preservation potential. Dinosaurs are fascinating and have a diverse fossil record, but we don’t have fossils of most dinosaur species because it’s relatively hard for species that lived on land to become fossilized, and bone is not always as durable as you might think. However, rhynchonelliform brachiopods make their shells out of calcium carbonate minerals which have a very high preservation potential, so they have a rich record. Additionally, the chemistry of the shells of rhynchonelliform brachiopods can tell us a lot about the environmental conditions at the time in which they lived.

BSJ: The Late Ordovician Mass Extinction (LOME) is thought to have been caused by a greenhouse-icehouse transition. What is the greenhouse-icehouse transition?

SF: Greenhouse and icehouse are paleoclimate shorthands for a very warm world and a relatively cold world. Right now we live in a very transitional time. It’s an icehouse climate state, since we still have major continental glaciers covering all of Antarctica and Greenland. But, as you are aware, we are busily leveraging ourselves onto the greenhouse climate spectrum. Greenhouse climates are ones where we typically have a high inventory of greenhouse gases—carbon dioxide and methane—and little ice at the poles. The most recent major greenhouse state was about 45 to 55 million years ago in the Eocene. From that time, we have fossils of alligators from Ellesmere island in Canada, which even then was above

the Arctic circle! Most of the Ordovician period is a relative greenhouse climate, but towards the end we see rapid climate change and growth of very large glaciers on the supercontinent of Gondwana. This coincides approximately with the extinction events.

BSJ: Prior to your research, why was the Ordovician period regarded as nonselective?

SF: What I mean by selectivity is the pattern of extinction versus survival across different groups of organisms. Understanding the cause of extinctions is hard because all we have is observations. We can’t do controlled experiments to see what drove brachiopods extinct. We have to rely on patterns and correlation. One of our main ways of getting insight into the causes of an extinction event is looking into its selectivity. Which groups and lineages went extinct, which ones survived, and how do they differ from one another? Are there patterns that can tell us about cause? If you think of some of the other major mass extinction events, there are conspicuous diverse groups of animals and plants that go entirely extinct. For example, in the Cretaceous-Paleogene mass extinction, the non-avian dinosaurs became completely extinct. In the LOME, we see very high extinction at low taxonomic levels—species or genera—but it’s distributed across most of the major groups of animals that existed at the time, and very few high-level groups of animals become entirely extinct. We don’t see a strong selective signature in terms of certain groups being driven to extinction and other groups persisting. Instead, we seem to see almost every group that existed at the time experiencing pretty high, but not total, losses.

BSJ: Your research shows the potential for the LOME to have been selective along certain axes.¹ How did you use predictors to quantify this selectivity?

SF: When we look at the distributions of genera that go extinct, we can see that there is a particular signature with respect to both geographic distribution and depth in the oceans. What we see for the LOME that differs markedly from extinctions that occurred before and after is strong selectivity with respect to the latitudinal distribution of species. Genera that had a wide distribution across latitudes did pretty well, and genera that had narrow latitudinal distributions experienced much higher extinction rates. That pattern is consistent with what we might expect if changing climate is a big part of what’s driving them extinct. As the climate cools or warms, the water masses at their habitual temperature range are going to shift—to the equator if it’s cooling, and to the poles if

“The Ordovician is an interesting period because it contains both a major diversification and a mass extinction event.”

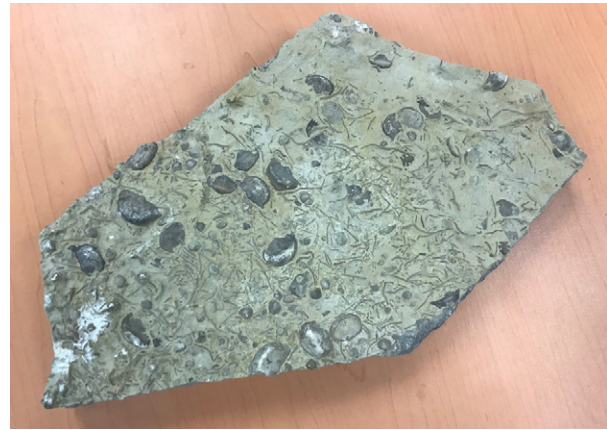


Figure 1 (left) and 2 (right): Fossils of Ordovician brachiopods in Dr. Finnegan's collection.

it's warming. And groups that were already widely distributed wouldn't be particularly sensitive to temperature because they have a wide thermal tolerance range. But narrowly distributed genera that have demonstrated a narrow temperature range are going to be in trouble. The other pattern we see is genera found exclusively in relatively deep water go extinct at much higher rates. That's likely because the distribution of dissolved oxygen correlates closely with depth. By looking at the chemistry of rocks during this interval we can determine there were big changes in the amount of dissolved oxygen in the oceans, suggesting this is also a major part of what caused the extinction.

BSJ: Could you go a little further into how the oxygenation affects the extinction of species at greater depths?

SF: It's a very active area of research and there has been a lot of back and forth in the literature over what's happening through this interval. Partly, it's because this happened almost half a billion years ago and it's very hard to reconstruct changes in local oxygen conditions. For a long time, our expectation was that when the climate cools, oxygenation of deeper waters will increase because cooler water holds more dissolved oxygen. The water at the poles, for example, has more dissolved oxygen than the water at the equator. One of the things we worry about now is that as the oceans warm they will also lose oxygen, adding an additional stress to organisms living there. For a long time, the thought was that as the climate cooled during the late Ordovician, deep water was oxygenated. Under most conditions, we wouldn't expect oxygenation to cause extinction, but if we have ecosystems that are adapted to low oxygen conditions, it may not be great for them. As we get better at reading the chemistry of the rocks, we are beginning to see that, at least in some places, the opposite happened and you got deoxygenation of relatively shallow waters. So, the whole community is in the process of working out how to combine the observations we are getting from geochemistry and extinction patterns with climate models to understand how this could all be happening at once.

BSJ: Could you explain what cratonic seaways are and how the LOME specifically affected the fauna there?

SF: A craton is a geological term for the stable, interior part of a continent. Cratonic seaways occur when you have extensive ocean flooding onto the continents. Because the LOME was a greenhouse climate state with very little continental ice and warm oceans, the continents were very extensively flooded. Most of the places where we now have good records, like North America, Northern Europe, Northern Africa, China, and Argentina were largely flooded by oceans at this time. But where we are now worried about flooding the continents as a consequence of global warming, in the Late Ordovician period flooded continents were great ecosystems for marine animals to live in. For example, these fossils (Fig. 2) come from a cratonic seaway that was in what is now Ohio and Kentucky—which in the Ordovician would have been a great place to go snorkeling. But as the ice sheets grew on Gondwana, sea levels dropped and cratonic seaways drained away, so some of the animals that lived in those seaways may not have been able to establish themselves in open marine ecosystems with very different environmental conditions, leading to greater extinction of fauna inhabiting cratonic seaways.

BSJ: You used similar predictors to the ones we talked about earlier to create a model identifying the most surprising victims in mass extinctions.² How were you able to determine unexpected victims by applying this model to the LOME?

SF: The idea here is pretty simple. Whenever we have a big extinction, certain groups of species that go extinct are going to help us understand the causes of extinction, and others not so much. Extinction is a fact of life. Any period of time we look at, there's always groups that go extinct and new groups that appear in the fossil record. So if we want to try and look at the pattern of extinction and figure out what causes mass extinction, we want to filter the surprising extinctions from the ones that are less surprising. The analogy I always use here is to epidemiology, where

we try to understand a major epidemic that occurred in the past. For example, most of the time epidemic influenza has a pretty distinct mortality pattern—mortality rates are higher in very young and very old people. In 1918, the “Spanish Flu” swept all over the planet and killed somewhere between 20 and 100 million people worldwide. And if you look at the distribution of mortality in the Spanish Flu compared to other flu epidemics, like other epidemics there was high mortality among the very young and the very old, but there was also a peak in mortality among healthy, young adults—people we normally consider to be least vulnerable to flu. That’s what you would refer to as an unexpected victim. This paper tried to do something analogous using extinction patterns in the 10 to 15 million years preceding the LOME to determine predictors for extinction.² And as with the flu example, they are not very surprising. If you are very narrowly distributed, you are usually pretty likely to go extinct at a given time. But there are also genera, such as the *Foliomena* Fauna, a distinctive group of brachiopods that lived in deep tropical oceans, where it’s very surprising to us that they go extinct. This is a major departure from the normal extinction regime.

BSJ: One factor that may have influenced your results was sampling bias. Can you explain the concept of sampling bias?

SF: An issue with the fossil record is that it’s incomplete in many different ways. So what we call sampling bias collapses a whole set of processes and events that multiply together to determine the likelihood of having a fossil record of any particular individual or species that existed at some point in the past. Brachiopods have nice mineralized hard parts that hang around after they die, but jellyfish, for example, will get fossilized only under exceptional circumstances. So, our sampling of brachiopods is much better than our sampling of jellyfish. Additionally, there are many places where I can’t see the Late Ordovician rocks because they are buried under younger rocks and the only way to look at them is by coring down. So another sampling bias is found anywhere the record of the organisms might exist but can’t be accessed. On top of that there is true sampling bias, which occurs when we haven’t sampled even the parts of the rock record that we can get to. Unsurprisingly, the most intensively sampled parts of the fossil record tend to be the ones located in wealthy industrialized countries where people have the luxury of spending their time studying fossils. There’s a strong bias towards North America, Northern Europe, and increasingly China, but there are still big parts of the world where we don’t know as much as we would like about the fossil record for socioeconomic or practical reasons. I’d like to know much more about the fossil record of Brazil, but I’m not going to advocate clearcutting the Amazonian rainforest just to study the fossil record better.

BSJ: You’ve used past extinction risk predictors to predict “intrinsic risk” in modern marine fauna.³ Could you explain the concept of intrinsic risk?

SF: In this paper, a couple of colleagues and I brought together both paleontologists and modern biologists who were interested in extinction in marine environments, and tried to think about how we can bridge the time gap between the fossil record and the modern day.³ The fossil record shows us that not all groups that exist in the oceans today are equally vulnerable to extinction under normal conditions. If we look through the last 23 million years of Earth history—the period of time during which the major groups that dominate marine ecosystems today were already in place—we can identify a set of relatively diverse groups of animals that have pretty good fossil records. We looked at molluscs, sea urchins and their relatives, corals, marine mammals, and sharks, and made a model for each of them determining the relationship between their ecology and aspects of their geographic distribution and extinction risk. Then we projected that on to the modern world (Fig. 3). The underlying idea is pretty simple: as we begin to worry about which genera will be most affected by the modern era of anthropogenic change, it is useful to identify genera that might be at intrinsically high risk of extinction anyway. You could argue that we don’t need to worry about groups that already have a high intrinsic risk, or you might say these are the ones we really want to focus conservation efforts on. That’s a policy and planning question. But the hope is that this model can serve as a kind of baseline to compare to our growing body of information about modern population response to climate change, ocean acidification, overfishing, deoxygenation, plastics, and all of the myriad awful things that we worry about in modern marine ecosystems.

BSJ: You found a high intrinsic risk for tropical genera. Was this more reflective of environmental characteristics or characteristics of the actual genera?

SF: We don’t really know. There are some periods of time where tropical genera seem to exhibit higher extinction risk, but not uniformly. In some cases, that’s associated with episodes of cooling, which may affect the tropics more strongly. But what we actually found is that simply being narrowly distributed is a big determinant of extinction risk, and that on average there are more narrowly distributed genera in the tropics than there are in extra-tropical regions. It also may be related to thermal tolerance range. Species that live at high latitudes experience a much larger range of temperature conditions in a given year than do those that live in the

“Naively, we think that cooling might be bad for tropical ecosystems, but warming might also be bad for them.”

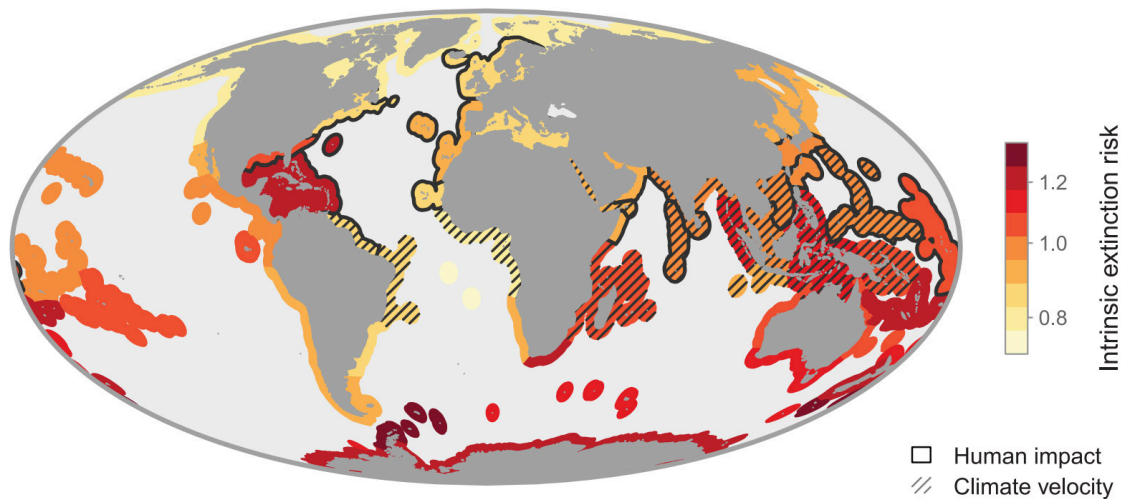


Figure 3: Overlap of global areas with high extinction intrinsic risk with global areas of high human impact.³

tropics. As the climate begins to shift, groups that have very narrow thermal tolerance ranges may be at much higher risk of extinction, depending on their adaptability. Also, because the solubility of oxygen is a function of temperature, some species in the tropics are already barely getting enough oxygen to function. For marine ectotherms (cold-blooded animals), warmer temperatures mean higher metabolic rates but at the same time less oxygen. So that's a bad combination. Naively, we think that cooling should be bad for tropical ecosystems, but warming might also be bad for them.

BSJ: How can the overlap of this intrinsic risk and our human impact help predict which coastal regions may face high extinctions in the future?

SF: This is absolutely the toughest question. This paper was part of a big working group involving both paleobiologists and marine biologists, and we wrestled with this question a lot.³ We're looking at very long time spans, and the question is: what's the time frame which we're actually planning for? Are we crafting policy for ecosystems a million years in the future or are we crafting policy for more immediate concerns? For example, if we look at the kinds of corals that exhibit high intrinsic risk in the record versus the kinds of corals that are currently thought to be at greatest risk, they're not generally the same groups. That tells us we might

need to be thinking about processes that play out on longer time spans than we can observe directly, or alternatively that the current anthropogenic impacts change the fitness landscape so much that the old rules no longer apply. This is one of the major challenges we face in studying evolution and ecology: we know that relevant time spans extend out to thousands and millions of years, but in most cases we only have direct observations of populations for maybe a few decades. There are important processes happening over longer timescales that we're missing when we only think about vulnerability in the short term.

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"We might need to be thinking about processes that play out on longer time spans than we can observe directly."

OPIATE ADDICTION AND ITS CONFOUNDING CRISIS

BY ASHLEY JOSHI



Whether taken recreationally or as prescribed, ingested opiates alter our brain chemistry and pave a road to possible addiction. While the opiate crisis as we know it today has only come to the forefront in the last decade, the effects of opiate abuse on our society can be traced back to the nineteenth century, when drugs such as morphine came into mainstream use. Due to modern technology and the mass production of drugs, opiates are incredibly accessible and widespread in their use today. Therefore, educating individuals about the issue at hand can prevent the crisis of opioid addiction from escalating.

RECEPTORS

The first step in addressing the problem of opiate addiction is understanding how opiates rewire the brain.¹ Acting on a stimulus from the environment, neurotransmitters—chemicals passed through nerves in the body—activate receptors in the brain to perform a function. Dopamine is a neurotransmitter that floods the brain as a direct result of opiate ingestion. This neurotransmitter rewards natural behaviors by producing ecstatic feelings, such as relaxation or intense joy. The over-activation of

reward circuits is what generates addiction: the mind is neurally rewired to seek the elation that is brought about by opiates.²

But rewiring brain chemistry has harmful repercussions. Dopamine activates the μ -opioid receptor (MOR), which triggers social interaction and decreases hunger. This neurotransmitter also activates the κ -opioid receptor (KOR), which triggers uneasiness or agitation—sensations that are far from ecstatic. Problems with KOR function can lead to psychiatric disorders such as psychosis, a mental disorder in which thoughts and emotions become completely disconnected from reality. Evidently, addiction distorts both behavior and perceptions of reality by disrupting the function of MOR and KOR. Medications for long-term treatment of addiction must address these repercussions.³

TREATMENT

The Food and Drug Administration (FDA) has approved three medications to alleviate the effects of opiate addiction: buprenorphine, naltrexone, and methadone. These drugs act on MOR and KOR receptors by producing or blocking a key physiological response of addiction. Buprenor-

phine, in particular, can produce and block a physiological response simultaneously.

In certain case studies, buprenorphine is a plausible medication for opiate addiction. Buprenorphine significantly lowered opiate and cocaine addiction in patients who had used these drugs for more than 10 years.⁴ Notably, opiate addicts who take buprenorphine can discontinue opiate use without experiencing the withdrawal symptoms that are typical of most other opiate-countering medications. In rhesus monkeys, researchers found that buprenorphine reduced self-administration of cocaine for up to 120 days.⁴ Although buprenorphine seems to be a viable medication to depress symptoms of addiction, scientists have yet

"The effects of opiate abuse on our society can be traced back to the nineteenth century, when drugs such as morphine came into mainstream use."

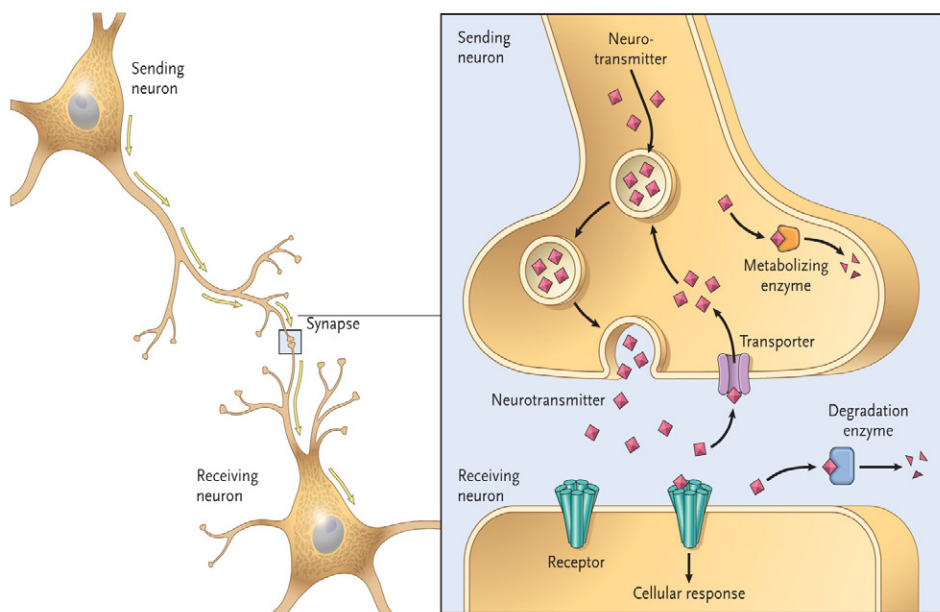


Figure 1: Neuron and receptor activity. The presynaptic neuron releases neurotransmitters. These neurotransmitters travel across the synaptic cleft and bind to receptors on the postsynaptic neuron. This process happens in the area between the anterior end and posterior end of each neuron in a gap called the synapse.

to conduct postexperimental studies to corroborate these findings.

Another hot drug on the market, naltrexone, can be compared to buprenorphine as a feasible treatment for addiction.⁵ Although buprenorphine activates MOR and KOR receptors, naltrexone works as an inhibitor, blocking MORs and KORs entirely. Hence, overdose or misuse of naltrexone results in severe withdrawal symptoms. Naltrexone needs only to be taken as a shot once a month, whereas buprenorphine must be taken as a pill daily. However, naltrexone is still two to three times more expensive than buprenorphine. It seems as though the benefits naltrexone offers do not compensate for its flaws. For these reasons, naltrexone, like buprenorphine, is another feasible yet inadequate medication for opi-

ate addiction.

Methadone, another drug approved by the FDA, behaves much like buprenorphine. In addition to producing minor addictive effects, methadone also produces milder withdrawal symptoms in patients. However, studies have found that consumers are more likely to abuse methadone as a prescription medication because it can be taken with addictive opiates without repercussions. In fact, many individuals take this drug with a regular dose of heroin.⁶ This phenomenon most likely accounts for the high mortality rates observed in methadone-prescribed patients in the United Kingdom during the 1990s.⁷ Despite its potential benefits, the misuse of methadone highlights its inadequacy in curing opiate addicts.

IMPLEMENTATION OF WHAT IS LEARNED FROM THE OPIATE CRISIS

Despite the available treatments, opiate addiction remains a crisis, as there is no single medication that can “solve” the epidemic.⁸ Still, research shows that individuals can reduce their likelihood of addiction in a number of ways, while other findings reveal that understanding the mechanisms

of addiction might be beneficial in treating other disorders.⁸

For instance, researchers at the University of California, Berkeley conducted behavioral experiments that demonstrated that engaging in mentally cognitive activities depresses the likelihood of addiction.⁹ Mice trained to search for cereal pieces in cups filled with wood shavings avoided chambers where they were given cocaine injections, preferring chambers where they were given saline injections. As a result of the training they received, mice were able to withstand chambers they sensed were harmful to their body. Linda Wilbrecht, a professor of psychology and neuroscience at Berkeley, determined that “learning opportunities may provide additional benefits, enhancing resilience in response to drugs with abuse potential.”⁹ The brain is rewired through cognitively stimulative training exercises and becomes less vulnerable to substances with abuse potential. Hence, engagement in simulating activities may save present and future generations from the opiate crisis.⁹

Research pertaining to the opiate crisis may benefit narcoleptic patients, even though it is currently unclear how opiate addicts should be treated. Prior research has established that opiate and heroin addicts have high levels of hypocretin cells that regulate arousal, wakefulness, and appetite. Jerry Siegel, a professor and chief

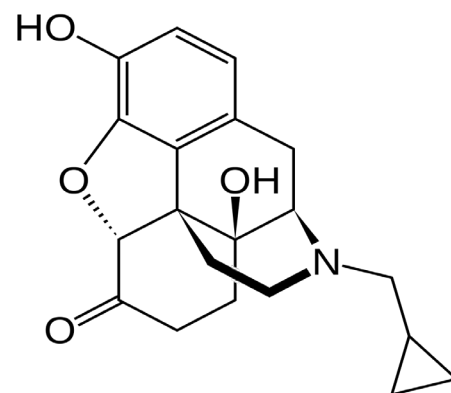


Figure 2: Chemical structure of FDA-approved medication naltrexone. This drug is used for the treatment of opiate addiction. As the body ingests opiates, it changes from its prior state.

“The brain is rewired through stimulative training exercises and becomes less vulnerable to addictive substances.”

"According to Dr. Siegel, the natural question to ask was whether opiates could treat narcolepsy."

of neurobiology research at the Brain Research Institute at the University of California, Los Angeles, led his team to apply this knowledge to studies involving narcoleptic patients: patients with generally low levels of hypocretin cells. Narcolepsy is a neurological disorder in which individuals frequently doze off during the day, although they sleep about the same number of hours as the average individual. According to Dr. Siegel, the natural question to ask was whether opiates could treat narcolepsy. By administering opiates to narcoleptic patients, the team found that patients had shorter episodes of dozing off during the day and increased levels of hypocretin cells.¹⁰ However, scientists have not yet explored the reverse: how treatments used for narcoleptic patients may potentially benefit opiate addicts.

While numerous research studies have provided insights into the crisis of opiate addiction, researchers have yet to determine a solution. Efforts to resolve this problem will require patience. But understanding the evolution of the opiate addiction crisis and assessing the current situation is a critical start.

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MYCOTOXINS IN DEVELOPING COUNTRIES: THE SILENT KILLER

BY ANDREA HE

E. coli. Pesticides. Salmonella. Mercury. These are dangerous food contaminants that in high quantities can cause detrimental health effects, not only in humans but also in animals. Food contamination has prompted protests, recalls, and removal of romaine lettuce, chicken, and fish from the shelves of grocery stores. However, mycotoxins are a relatively unknown type of food contaminant despite their global effects and their major impacts on developing countries. What steps should citizens take in order to improve human health?

THE HISTORY OF MYCOTOXINS

Mycotoxins are secondary metabolites produced by various types of mold and fungi and can cause a disease known as mycotoxicosis. The effects of mycotoxins range from mild to severe and are often specific to the organism by which they are produced. For example, aflatoxins, generated by the *Aspergillus* species of fungi, have carcino-

genic effects.¹ Meanwhile, fusarial toxins from the *Fusarium* species can cause nervous system damage in horses and cancer in rats.² Humans most often ingest mycotoxins through foods such as cereal grains, milk, and meat. Due to livestock consumption of mold-contaminated foods, beef is a common source of mycotoxins in the human diet.³

Human interaction with mycotoxins has a lengthy history. Over 10,000 years ago, humans transitioned from a lifestyle of hunting and gathering to one of organized agriculture. This change necessitated the storage of food, especially grain, for longer periods of time.⁴ Mold would strike when the grain rested in caves. Thus, to prevent mold, grain was stored on raised platforms or in silos where mold was less likely to reach it. In 800 A.D., during the Roman Empire, the government documented a case of what now appears to be mycotoxin poisoning, most likely caused by consuming rye contaminated with the mycotoxin ergot. Ergot can cause headache, diarrhea,

and gangrene. Mycotoxin diseases such as this one may have been transmittable from a pregnant mother to her infant.⁴ However, at the time, the Romans referred to the disease as “slow nervous fever.” The name of the disease did not reveal the role of mycotoxins, as people did not realize these poisons existed.⁴ These interactions between humans and mycotoxins are still present today.

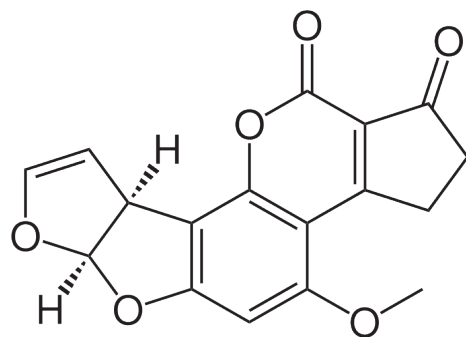


Figure 1: Chemical structure of aflatoxin B1.

THE PROBLEM OF MODERN DAY MYCOTOXINS

Unfortunately, mycotoxins contaminate a large percentage of the world's food, especially grains. The Food and Agriculture Organization (FAO) of the United Nations has determined that 25% of the world's grain has been contaminated. This contamination can occur at many levels of production, from pre-harvesting to drying and storage.⁵ Many countries have adopted methods to prevent contamination by applying innovative techniques, such as crop rotation, irrigation, and pesticide applications. For example, by shelling corn when it is harvested and improving grain storage techniques to minimize moisture, the United States has maintained extremely low mycotoxin levels in the nation's corn supply.⁶ However, mycotoxins are still a prevalent problem in developing countries due to damaged and therefore inedible crops.⁴ Improperly stored food increases susceptibility to mycotoxins, which drastically reduce the amount of food available to people.

DISPROPORTIONATE EFFECTS ON DEVELOPING COUNTRIES

Mycotoxin contamination disproportionately targets developing countries. Due to globalization and industrialization, many impoverished populations are growing less diverse crops, with a shift towards refined grain consumption. Refined grains have a higher susceptibility to mycotoxins than unrefined grains such as sorghum and cassava, both of which were previously consumed by residents of these countries. As a consequence of this change in the production of staple crops, citizens of developing



Figure 2: Corn affected by Aspergillus flavus fungus.

regions in East Africa have higher risks of contracting mycotoxin-caused health problems. For example, researchers have noted similarities in the symptoms of mycotoxin contamination and autism. In humans, such symptoms include oxidative stress, inflammation, and intestinal permeability. These surprising findings are what encouraged scientists to investigate the effects of mycotoxins on the manifestation of autism.⁷ By analyzing the role of ochratoxin A (OTA), a specific type of mycotoxin, researchers discovered that OTA is involved in the regulation of autism-related genes. Diets that are low in OTA and high in probiotics could ameliorate autistic symptoms in patients that are OTA positive.⁷ Through findings like these, mycotoxins may shed light on a variety of other conditions. Mycotoxins can also impede the growth and development of children. This problem ultimately creates severe economic difficulties for countries such as India and Nigeria.⁴

In the developing country of Zimbabwe, further research is necessary to understand and address the problems that mycotoxins pose. Maize, which makes up 70% of the diet of Zimbabweans, commonly contains mycotoxins. Typical cooking methods in Zimbabwe do not reach high enough temperatures to kill the cells and spores within the food. Although the WHO and FAO of the United Nations have established upper limits for mycotoxin presence in food, relaxed regulations, droughts, and low fund-

ing often result in the failure of developing nations to observe and abide by these limits.⁸ Consuming food with high mycotoxin levels is an ineffective means of resolving food shortages caused by droughts and poses serious long term health problems for the populations in developing countries.

ADDRESSING THE PROBLEM

In order to combat mycotoxin contamination in both developed and developing countries, investigators need to conduct more research. Limited funds and poor analytical equipment generally prevent journals from publishing papers by researchers in Zimbabwe.⁹ Additionally, many researchers in developing countries are unable to afford the costs of publishing and conducting research.⁸ Beyond research, policies that address the needs of developing nations need to be proposed and passed. Mycotoxin levels vary by country, and in order to optimize regulations and policies, countries need accurate data regarding mycotoxin levels. This kind of information can be difficult to obtain, especially since other issues such as vaccinations and HIV testing often take precedence. For this reason, publicizing the dangers of mycotoxins is important in allowing people to realize and address their impact.

Raising public awareness surrounding mycotoxin contamination through public health campaigns and social media may

"Mycotoxin levels vary by country, and in order to optimize regulations and policies, countries need to have accurate data regarding mycotoxin levels."

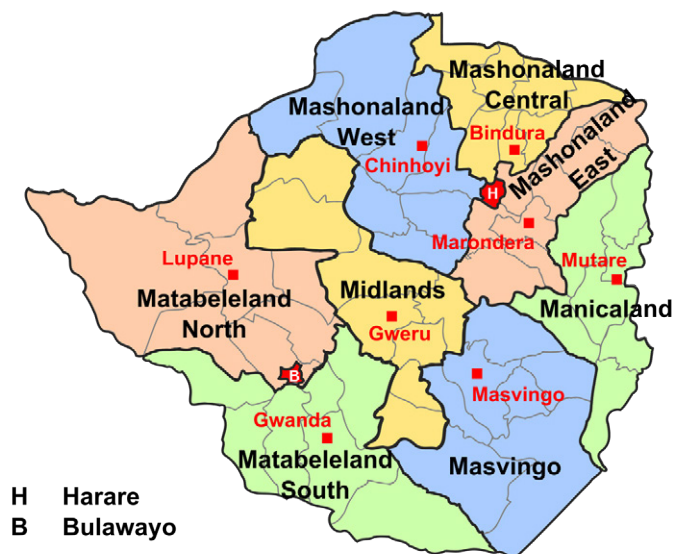


Figure 3: Map of Zimbabwe.

encourage the allocation of more funds to mycotoxin research. Furthermore, there is a strong need for countries to implement and enforce policies regarding mycotoxin contamination at the federal level. Model systems such as Good Agricultural Practices, Good Manufacturing Practices, and Good Hygienic Practices are researched and tested methods that can prevent mold and are a good starting points for policy development.³ Hence, the combination of research and policy offers viable steps towards the prevention of food waste and the health of future generations.

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IMAGE REFERENCES

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SCIENCE FROM THE BOTTOM UP: MOSQUITO-BORNE DISEASES IN NICARAGUA

Interview with Professor Eva Harris

BY MATT COLBERT, CASSIDY HARDIN, MELANIE RUSSO, KAELA SEIERSEN, AND NIKHIL CHARI

Eva Harris is a Professor of Infectious Diseases and Director of the Center for Global Public Health at UC Berkeley. Her research focuses on mosquito-borne viral diseases including dengue, Zika, and chikungunya in Latin American countries. We chatted with Dr. Harris about the cross-reactive relationships between Zika and dengue antibodies and the potential for certain concentrations of antibodies to enhance disease. But before we even got to our questions, Dr. Harris wanted to share with us what inspired her to start her research program and nonprofit organization in Nicaragua.



Professor Eva Harris.

EH: Can I just launch right in? I was always interested in science—I did my undergrad at Harvard, and then I came here to UC Berkeley for my PhD in Molecular and Cell Biology. I was at Harvard in the Reagan eighties during the Iran-Contra scandal, and I was very politically active. I wanted to connect politics and science in my career, but at the time there was no way to do this because this was way before global health was a concept. I had decided to go to Berkeley for graduate school, but I postponed that and went to Nicaragua because there was a revolution and I wanted to be part of it. I landed with my pipettes and everyone was unsure what to do with me. I was completely unprepared because I had never been “south of the border.” I had traveled very widely but I had never wanted to go to the developing world as a tourist. I wanted to contribute

something—but when you’re twelve and thirteen you don’t have much to contribute. At twenty I still didn’t have much to contribute, but I invented as I went along. It eventually became a thirty-year long, multi-million dollar program with hundreds of local workers. It has made a big impact on science in Nicaragua, and it has also become the basis for a large part of our non-profit, the Sustainable Sciences Institute (SSI), which focuses on building scientific capacity in developing countries worldwide. My vision is about doing good science but simultaneously connecting it in a way that makes the world a better place—I’m still an idealist. The vision has never been to have a top-down, vertical, North-South approach but to be more horizontal. Our goal is to directly address problems that are local priorities by applying methodologies in a way that is knowledge-based

and builds from the bottom up. Upon completing my PhD at Berkeley, I wanted to become a bridge between academia and health problems around the world, which eventually became known as global health. But I was doing this fifteen years before the term “global health” was even coined! I received the MacArthur “Genius” Award in 1997, which I used to start the nonprofit SSI while I was building my academic career here at Berkeley, and having a baby. When I came here I was interviewed by a program called Conversations with History. They asked me what I was doing and what it was called, and I wasn’t sure what to name it, so I told them International Science... That was my moment to coin “global health,” but I didn’t! In many ways now, the moniker “global health” has become a way for US universities to have glitzy programs which are less about the welfare of their partners than their own universities. However, that is not what I espouse. I accepted a professorship in Infectious Diseases at Berkeley, which became a platform for me to have the independence I needed to build what I wanted to build. Of course, you pay in academia because you don’t really sleep for the next fifty years. I didn’t have any experience in dengue, virology, immunology, statistics—none of what I ended up working on! I had my PhD in yeast genetics and I’ve basically winged it for the rest of my life. I’ve always thought science as a horse, and I’m just hanging onto the tail, trying to learn as much as I can along the way. After accepting the professorship, I was working on multiple infectious diseases in several different countries. People told me to focus, but I wasn’t sure how to because I was interested in everything. Gradually, I forced myself to choose vector-borne diseases and settled on dengue because it was a priority disease in every country in Latin America I worked in. The dengue virus is interesting—it’s kind of like a breathing ball (Fig. 1)—and there were many unanswered questions about it and surprisingly little research at the time. We also couldn’t use existing animal models because den-

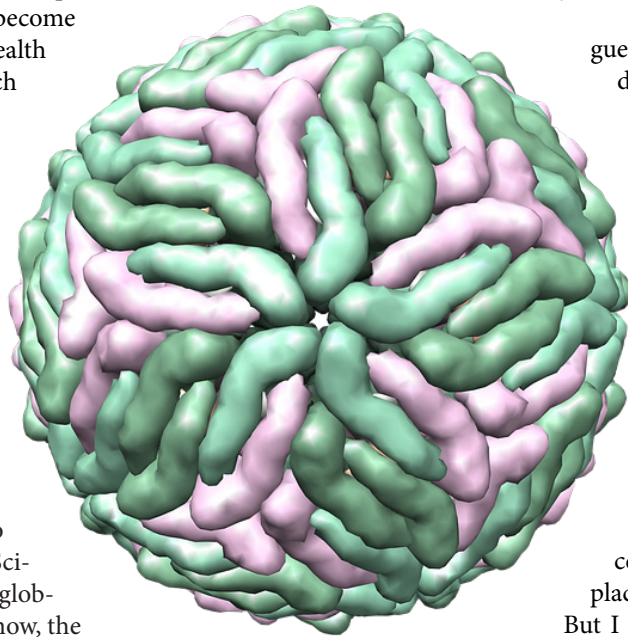


Figure 1: Dengue virus 3D structure.¹

gue doesn’t occur in mice the same way it does in humans. So I established a broad program that spans virology, pathogenesis, and immunology, which my lab studies here at Berkeley, to epidemiology, diagnostics, clinical aspects and control, which we study in close collaboration with my colleagues in Nicaragua. When Zika came along, it expanded everything, since we were able to port laterally from dengue across numerous disciplines. We began about forty new projects in two months! There were so many more questions—pregnancy, microcephaly, which cells does it invade in the placenta, diagnostics, cross-reactivity, etc. But I still continued doing dengue research because that is the focus of my grants. I was able to add supplements to expand all our work into Zika, and we were able to add Zika to all of the studies we had ongoing in Nicaragua, as well as add new studies of pregnant women. So that gives you a little bit of context.

BSJ: Let’s talk about your research on Zika first. What does it mean to be seropositive?

EH: Seropositive just means that you have been exposed to the pathogen of interest and therefore your body has developed antibodies to that pathogen, which we can measure. In this case, it is tricky because dengue and Zika viruses are very closely related antigenically, and there is a lot of antibody cross-reactivity. All the standard methods we had for serologically detecting dengue virus infection were now criss-crossed with Zika virus infection. However, using fifteen years of samples from patients with dengue and Zika, we immediately developed a sensitive and Zika-specific assay. This was the Zika NS1 BOB ELISA, a blockade of binding (BOB) ELISA that is based on a viral protein that is secreted from infected cells called non-structural protein 1 (NS1). A lot of people were using NS1 as an antigen, but both dengue and Zika antibodies can recognize Zika NS1. We worked with a small company that had developed human monoclonal antibodies to Zika virus. A monoclonal antibody is produced by a single B cell (a white blood cell that secretes antibodies) that is fused to make an immortal cell. The resulting hybridoma only secretes antibodies of that particular clonal lineage, that in this case recognize a specific site on Zika virus NS1. We label that antibody with an enzyme (which can be detected colorimetrically) and then compete it against the antibodies in patients’ sera. So, if you have had Zika, you will also have antibodies against that one Zika-specific site that can displace the labeled monoclonal antibody, reducing the color measured in the assay. But if you have dengue, you won’t have an antibody to that

“Our goal is to address local priorities by applying methodologies in a way that is knowledge-based and builds from the bottom up.”

Zika-specific site, so you maintain the labelled antibody and the color. In this way, we can distinguish individuals who have been exposed and have developed antibodies to Zika from those who have developed antibodies to dengue.

BSJ: Your study sample was divided into three groups: pediatric, adult, and family. Across those three groups what factors had the largest influence on Zika seroprevalence (level of Zika pathogens exhibited within a population)?

EH: You would think that when a new pathogen is introduced into a naïve population, everyone would be equally exposed. But that was not the case, and we observed differences in Zika seroprevalence risk across both age and sex. We saw that females were slightly more exposed. In our original pediatric cohort of 3,700 children we observed an interesting linear rise in seroprevalence over age, but we wanted to examine not only how kids ages two to fourteen are impacted, but how the adults are infected as well. So we expanded our study to our household cohort, and we observed that Zika seroprevalence risk was more flat across ages in adults compared to the children's cohort. We then compared seroprevalence across the whole household study—kids and adults

“In a multivariate model, body surface area was revealed as the sole significant risk factor for getting Zika infections.”

(Fig. 2). As before, we observed Zika seroprevalence risk increased with age and then flattened out. We were wondering why this was, and we noticed that kids who were obese or overweight had slightly higher seroprevalence. Because of that, we started looking at body mass index (BMI), as well as what turned out to be the best correlate: body surface area (BSA). When you examine variables one by one in a univariate analysis, both age and BSA are significant. Another significant variable is school session. Children in Nicaragua go to school either in the morning or the afternoon. The kids that went in the afternoon were also at a higher risk of seroprevalence. However, in a multivariate model, BSA was revealed as the greatest significant risk factor for getting Zika infections.

Then we realized, not only do children get bigger as they get older, but the afternoon session was when the older kids went to school. Everything was collinear with body size. Additionally, mosquitoes are attracted to carbon dioxide. If you're bigger, you breathe out more carbon dioxide. Women sometimes breathe more rapidly than men, and when you are overweight or obese, you can also breathe more rapidly. We also added seroprevalence to our spatial analysis, because every child in our cohort study has a GPS point for their house. If you notice, in Fig. 3, the purple is clustered at the western end of our study site, around the cemetery. We then went to the cemetery and measured all the mosquito breeding sites there. *Aedes aegypti*, the biggest mosquito carrier of Zika and dengue viruses, breeds in clean water around people's homes. In other community-based projects, we explain to people why they should clean up standing water around their homes—but no one is doing that in a cemetery. In fact, the cheapest tombstone one can buy in the cemetery is a cross with two little holders for flowers on either side. These flower-holders fill with rainwater—even if you are not bringing water for the flowers, these holders are collecting water anyway.

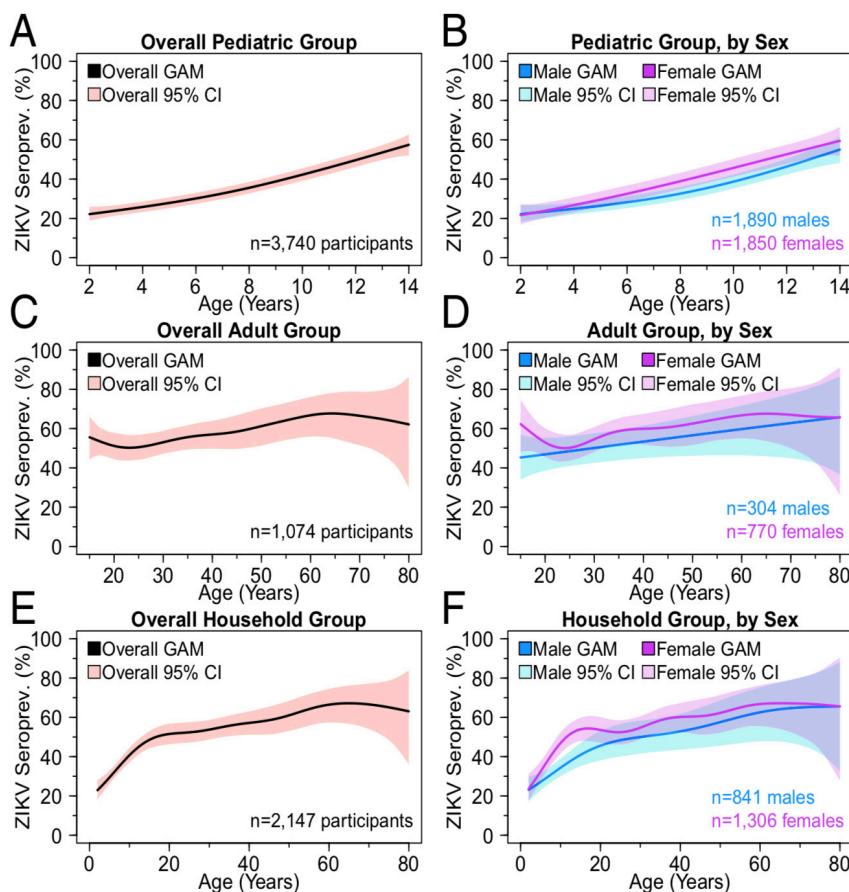


Figure 2: Zika seroprevalence in pediatric, adult, and household cohorts.¹

Additionally, some crypts are broken and water can seep in, which makes wonderful mosquito breeding grounds.

BSJ: Could you define neutralizing antibodies and explain the concept of cross-neutralization?

EH: For measuring neutralizing antibodies, there is a different kind of assay consisting of cells, antibodies, and a virus. If the antibody binds to the virus in a way that blocks it from infecting the cell, then the infection is neutralized. One can measure infection either by a plaque assay or by flow cytometry using a labelled antibody that will essentially color a cell upon infection. Then one can perform a dilution series of the serum or monoclonal antibody in question and add that dilution series to the virus. If the antibody neutralizes infection, then at high concentration no plaques, or no colored cells, are obtained. As the serum or antibody is diluted, more and more plaques are obtained. We can measure the presence of neutralizing antibodies using an NT50 value, or neutralizing titer 50—the concentration of serum or monoclonal antibody that reduces the amount of plaques or colored cells by 50%. That value can be used to compare antibody neutralizing potency. Dengue is caused by four different virus serotypes, and antibodies can cross-react with these serotypes and some can cross-neutralize different serotypes. Other antibodies are type-specific—these are powerful and will protect you from future disease upon infection with the same serotype. Anti-Zika virus antibodies are not only cross-reactive with dengue viruses, they can be cross-neutralizing to some extent as well. So one question was: even though there is cross-neutralization, could we still use these methods to distinguish dengue from Zika? If done properly, we can.

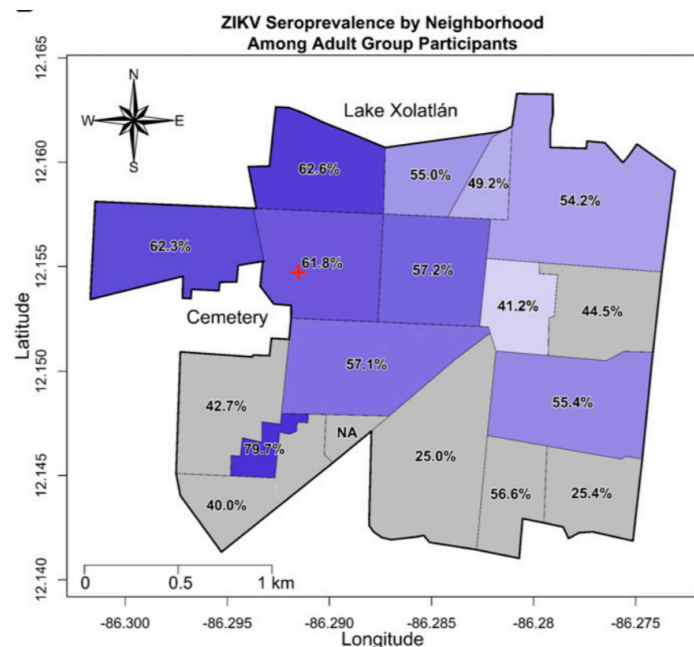


Figure 3: Spatial distribution of Zika seroprevalence.¹

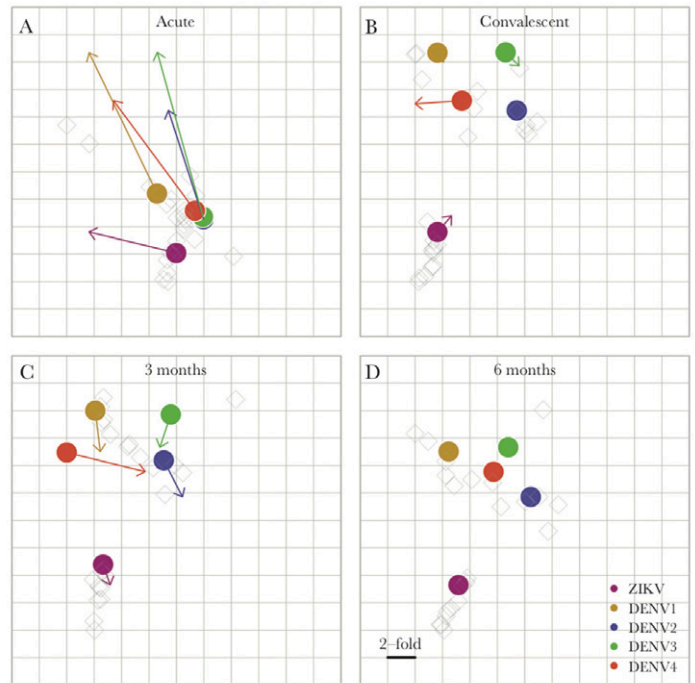


Figure 4: Antigenic map of dengue 1-4 and Zika viruses.²

BSJ: Does cross-neutralization of Zika occur in dengue-immune individuals? What does that tell us about Zika's presence in the dengue serocomplex?

EH: The short answer is yes; you can have cross-reactivity and cross-neutralization. The question is: what is the magnitude? If you've had a dengue virus (DENV) infection in the past, you will have higher neutralization to dengue virus in the future. Cross-neutralization of Zika virus (ZIKV) also occurs in dengue-immune individuals, but on a much smaller scale than it does for other dengue serotypes. Then we asked the opposite question: if you've had Zika, do you cross-neutralize DENV? You do, but again, at a much lower level than ZIKV. We then made an antigenic cartography map (Fig. 4) where we plot distance as a function of NAb titers. We essentially plot each virus as a ball in three-dimensional space. When you collapse that into two dimensions, you can see where those balls are in relation to each other, and what we found is that early after infection, ZIKV was in the same region as dengue—that's why people thought they were very similar. But as time went on, ZIKV really pulled away from the dengue viruses on this antigenic map and therefore we believe Zika virus is in a distinct serocomplex from dengue viruses. I direct a big grant that brings together academic groups from around the country to investigate adaptive immunity to dengue and Zika. Dr. Aravinda de Silva's group at the University of North Carolina, Chapel Hill, has developed a method for pulling out certain subsets of antibodies, for instance, antibodies that recognize DENV 2 (one of four DENV

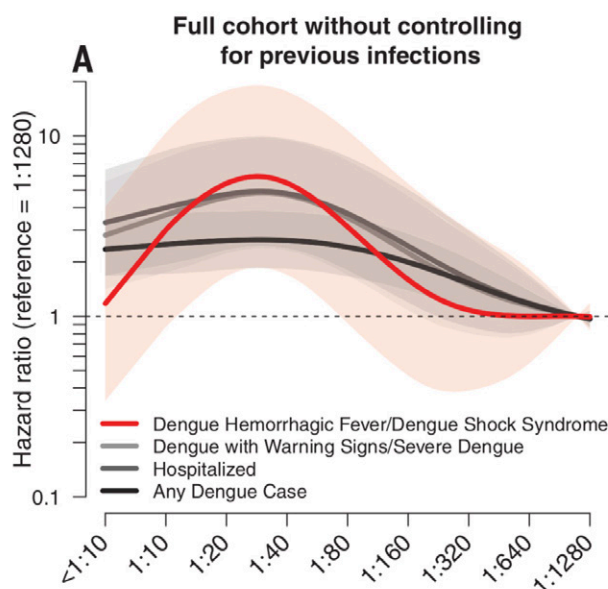


Figure 5: Dengue antibody titers illustrate a peak enhancement between 1:21 and 1:80 NAb ratios.³

serotypes). This allows us to study a polyclonal mix of antibodies but remove all the cross-reactive antibodies and be left with just the type-specific antibodies. Using this method, we found that in both travelers and endemic populations, even though there are a ton of dengue and Zika virus cross-reactive antibodies, they're really not contributing to the Zika neutralizing antibody titer. What's really contributing are Zika type-specific antibodies. In other words, dengue antibodies are not necessarily cross-neutralizing to ZIKV even if they are cross-reactive.

BSJ: Given all the research with the cross-neutralization between dengue and Zika, what would you say to claims about the potential of single possible vaccine for both Zika and dengue?

EH: Initially, before we did that last set of experiments, we thought that the presence of cross-reactive antibodies against both dengue and Zika was a positive sign for a single vaccine. But the fact that the most potent Zika neutralizing antibodies are Zika-specific means that you couldn't have a dengue vaccine that would work against Zika. What people are working on, since the dengue vaccine has four different serotypes, is adding Zika as a fifth virus. A dengue vaccine isn't going to work against Zika, but both viruses could potentially be included in one vaccine.

BSJ: We've spent a lot of time talking about neutralizing antibodies, but at some concentrations antibodies can enhance disease. Can you explain the concept of antibody-dependent enhancement (ADE)?

EH: Antibody-dependent enhancement is a concept that's been around for a really long time, and it has to do with

the fact that there are multiple ways for a virus to enter certain target immune cells. One is through what we call a cognate receptor, which is essentially a receptor that recognizes the virus and brings it into the cell. But you can also have antibodies to that virus that recognize the virus but don't actually neutralize it, as we discussed above. This causes an immune complex where that virus is still alive even though it's bound to an antibody. The constant region (Fc) of that antibody interacts with Fc receptors on the target cell surface, which bring the antibody and the live virus into the cell. So there are two routes into an Fc receptor-bearing cell: one through the cognate receptor, and another through the Fc receptor. In general, the Fc receptor route is only supposed to bring in dead or neutralized viruses, but if you have an antibody that has not neutralized your virus, it gives your virus a "stealth" way of entering the cell. This way, the virus does not trigger the innate immune response within that cell. Having this extra route ends up increasing the infection of that immune cell, which then activates T cells, which secrete cytokines. Then a cytokine "storm" is created that leads to pathogenesis. In the paper you are referring to, we didn't actually show ADE by this mechanism, but we did show antibody-enhanced disease in human populations, meaning that there is an increased risk for severe disease in people with a specific concentration of pre-existing antibody than those with more or less of that antibody.

BSJ: In this study you showed that subjects with antibody ratios between 1:21 and 1:80 were at a significantly larger risk for dengue hemorrhagic fever/dengue shock syndrome (DHF/DSS)—the most severe cases of dengue symptoms. How does this ratio relate to the concepts of antibody-dependent enhancement and antibody-enhanced disease?

EH: At this point, we have evidence to show that certain concentrations of antibodies cause antibody-enhanced disease, which is a non-mechanistic immune correlate. Theoretically, we could be seeing ADE, which is a mechanistic correlate. The idea is if you have no antibodies, the virus is only getting into the cell via the cognate receptor. If you have many, many antibodies, even if they're not great, they fully coat the virus so it can't get into the cell. But if you have antibodies that are not great, and you don't have enough of them to fully coat or neutralize the virus, they actually help the disease by forming an immune complex and allowing the virus to enter the cell via the Fc receptor. That's why we observed a greater risk of DHF/DSS at that particular range of antibody concentration. There has been a huge controversy in the field over how to measure enhancing antibodies. It's generally

"The dengue vaccine that's been licensed actually causes ADE in dengue-naïve people."

done in vitro by taking a cell line with only Fc receptors—no cognate receptors. You treat these cells with DENV and get no infection until you add antibodies. Then with a dilution series, you obtain a curve similar to the one in Fig. 5 because there's no other way into the cell unless a certain amount of antibodies enable the virus to enter the cell via the Fc receptor route. The question is: what is that level of antibodies in a human? We avoided a lot of this controversy because we weren't testing any assay in vitro, we were just observing the natural antibody titers of children with disease.

BSJ: What implications does your research in ADE have on dengue vaccination?

EH: Big—because the dengue vaccine that has been licensed actually can cause ADE in some dengue-naïve children. A lot of us saw this coming and warned that ADE could be an outcome. But the company went ahead, and children were vaccinated in the Philippines, and it turns out that there are reports of more severe disease in dengue-naïve vaccinated children, which are currently being investigated. The company has changed its label to only recommend vaccination in dengue-immune individuals. The fact that we showed ADE can occur in humans concomitantly with the company's change in its label was a big deal. Now that vaccine can no longer be used in dengue-naïve children, so this study had a big impact.

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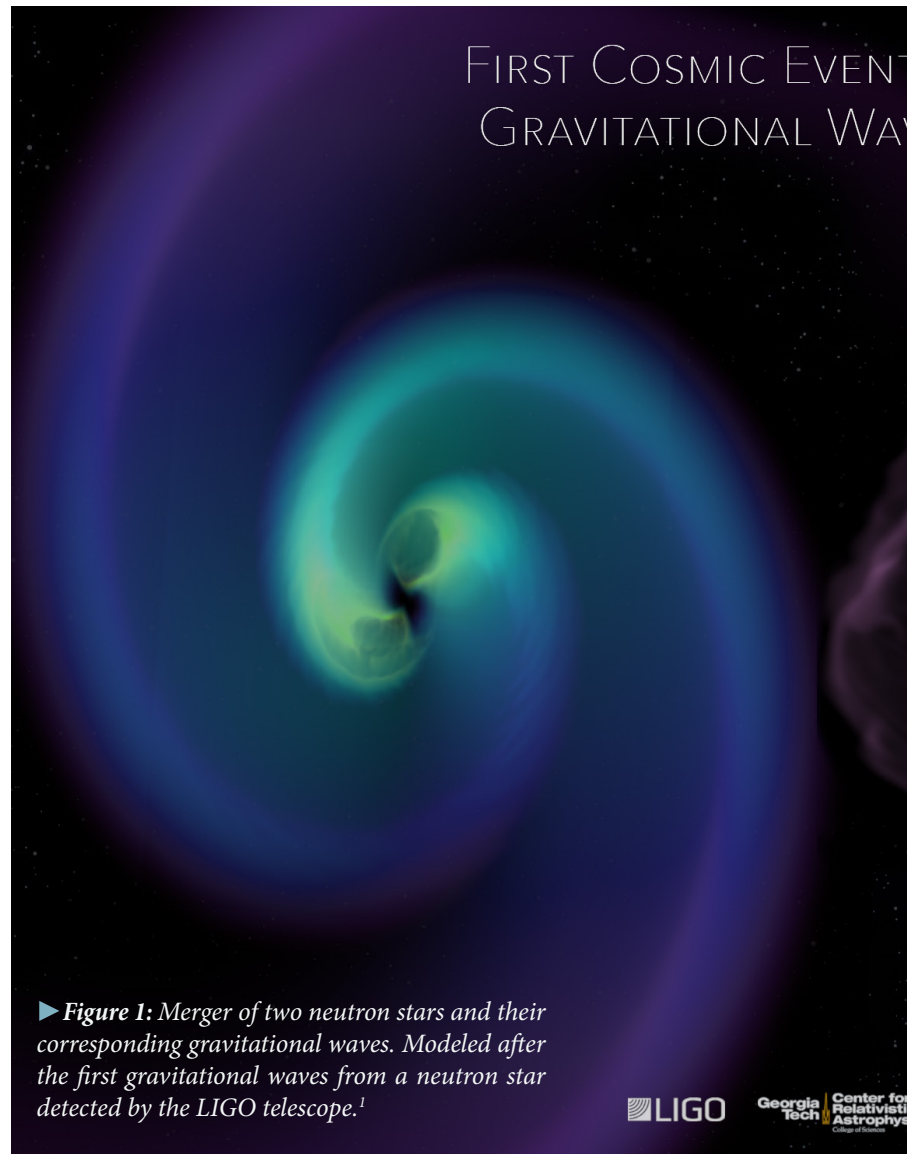
MEASURING THE UNKNOWN FORCES THAT DRIVE NEUTRON STAR MERGERS

*Interview with Professor
Eliot Quataert*

BY CASSIDY HARDIN, MICHELLE
LEE, AND KAELA SEIERSEN

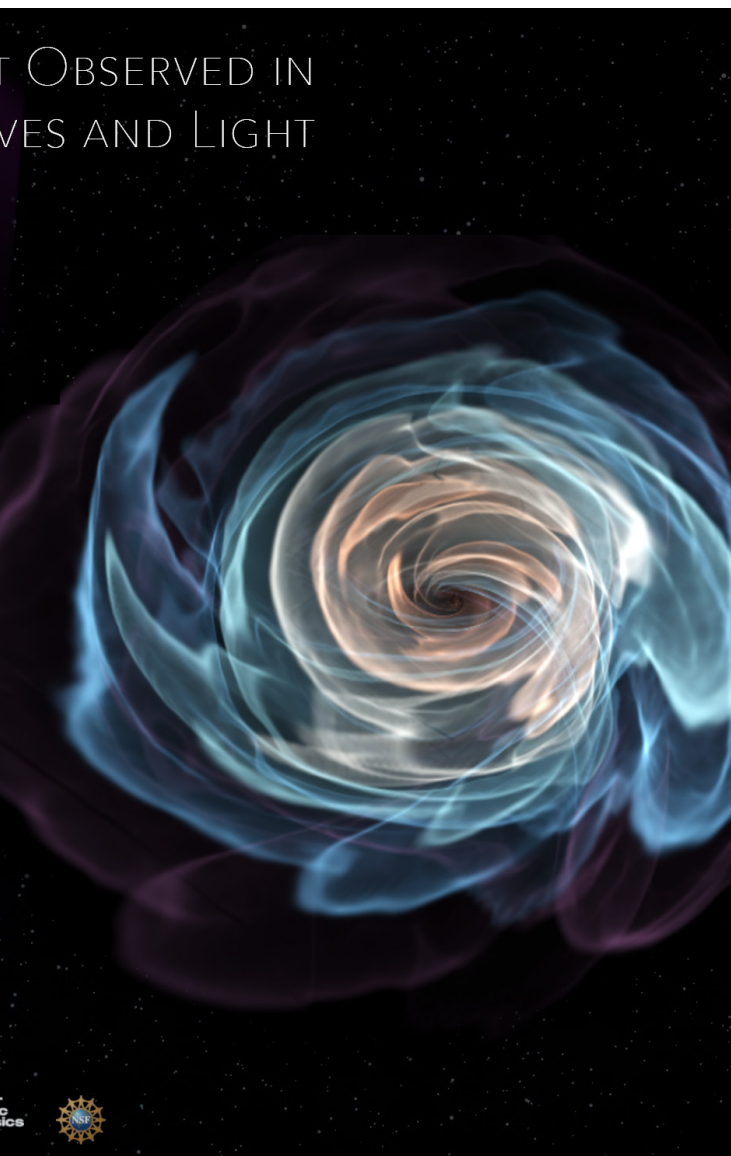


Professor Eliot Quataert.



► **Figure 1:** Merger of two neutron stars and their corresponding gravitational waves. Modeled after the first gravitational waves from a neutron star detected by the LIGO telescope.¹

Eliot Quataert is a Professor of Astronomy and Physics at the University of California, Berkeley. He is also the director of the Theoretical Astrophysics Center, examining cosmology, planetary dynamics, the interstellar medium, and star and planet formation. Professor Quataert's specific interests include black holes, stellar physics, and galaxy formation. In this interview, we discuss the formation of neutron stars, the detection of neutron star mergers, and the general-relativistic magnetohydrodynamic (GRMHD) model that was used to predict the behavior of these mergers. Analysis of these cosmic events is significant because it sheds light on the origins of the heavier elements that make up our universe.



now. It was really getting involved with research that allowed me to realize that I love astrophysics. One of the great things about astrophysics relative to particle or string theory is the close connection with observation. This interplay between the abstract, theoretical things I do and the observational side is really fun and exciting, and it also keeps my research grounded in reality. I think this combination is what really convinced me to do astrophysics in graduate school. Right now, I'm working on relating stars with black holes and investigating how stars collapse at the end of their lives. In some cases, if the exploding star forms a black hole, the surrounding material can form a disk around it, which can do all kinds of interesting things. I am trying to study this process.

BSJ: What are neutron stars and how do we observe them?

EQ: Neutron stars are the smallest stars we know of that we can still call normal stars, as opposed to black holes, which are smaller and weirder. Neutron stars consist of materials about the mass of the sun that have been condensed to the size of the Bay Area—roughly 10 kilometers in size. They are extraordinarily dense because you have all this material in a very small region. Under these conditions, the protons and electrons that make up normal matter are forced to combine into neutrons. This is why the matter does not end up as the standard elements we are familiar with such as hydrogen or helium. Rather, these stars are big balls of mostly neutrons, and the conditions in the star resemble those in an atomic nucleus, where the neutrons are packed close together. We observe neutron stars mostly through their light. They emit radio light in a clock-like manner called radio pulsars. We have observed thousands of these in our galaxy. Just recently, we were able to observe neutron stars in gravitational waves as opposed to normal light. When two neutron stars get close and spiral around each other, they alter gravity in time. This information that gravity is changing in time goes out into space in the form of waves, which Einstein had predicted, called gravitational waves. A telescope—the Laser Interferometer Gravitational-Wave Observatory (LIGO)—was able to detect the merger of two neutron stars through the measurement of gravitational waves that were created in the final 10 seconds of the merger.

BSJ: Could you tell us more about how neutron stars merge?

EQ: The Earth orbits around the sun, but it does not fall into the sun. In the case of two neutron stars orbiting close to each other, the gravitational waves take energy out of the system and cause the two neutron stars to move closer to each other. You can think of it loosely as gravitational friction, where friction causes things to slow down. In this case, friction causes the orbit to slowly spiral in. The two neutron stars slowly get closer to each other until they eventually become a single star. Although we are

BSJ: What originally interested you in astrophysics? Why did you start studying black holes and stellar evolution?

EQ: I was interested in physics and math as a high school student, and I was also drawn to more abstract things—I was not much of a tinker. I grew up in the country and was interested in photography, so I did a lot of night sky photography, and I think that was partially what got me interested in astronomy. When I was an undergraduate at MIT, I thought I wanted to study physics, and there wasn't a separate department for astronomy. I was really interested in doing research, and the first project I worked on was in an experimental lab. I hated it and did not think I was very good at it, so I knew I wanted to find a theoretical project next. My first theoretical project was studying sound waves of the sun, which for the most part is very different from what I do

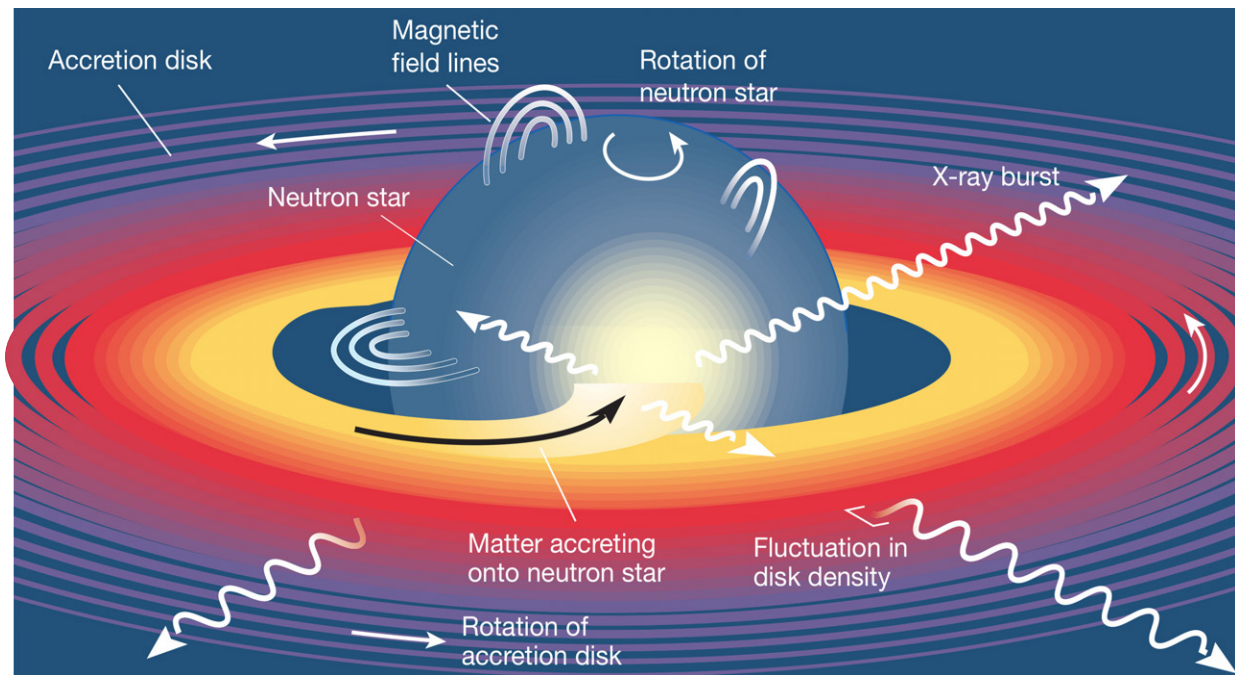


Figure 2: Annotated diagram of a neutron star and its surroundings.³ When a massive star dies and collapses after a supernova, the core is unable to withstand its massive gravity and all the atoms lose their structure, forming a neutron star. During this formation, the star has a very high spin and matter flies out, creating an accretion disk. This diagram portrays the behaviors of a neutron star and what happens around it.

not sure yet, this star likely collapses to form a black hole. We've used telescopes on Earth to measure the gravitational waves, and over time we observe these waves getting stronger and stronger until they eventually disappear. The waves disappear when the two neutron stars merge, forming a new object that sits there in space, not producing any gravitational waves.

BSJ: When two neutron stars collide, remnants fly out and create accretion disks surrounding the stars. What is the importance of studying the remnants of these mergers?

EQ: In general, accretion disks are a way of producing light and energy. You have a central object and you have stuff orbiting around it. If the central object is a neutron star or a black hole, the matter that orbits around it is moving really fast. As this matter moves around the central object, it gets really hot and can produce a lot of light and material that is flung out into space. The importance of accretion disks in general is that they produce some of the brightest sources of light we are able to see. In this recent case, we think the collision between two neutron stars produced what is likely a black hole with some sort of disk around it. This disk then flung some material out into space, creating and releasing many heavy elements. Initially, this material was mostly neutrons because they are the main component of neutron stars. As the material was flung out into space, the neutrons and the few protons that were around started to combine with each other to form heavier nuclei. We think this event produces elements like

gold, platinum, uranium, and some of the rare, unusual heavy elements in the periodic table that we haven't really found the origin of in nature. This event was sort of a confirmation that those elements could be produced through the collisions of neutron stars.

BSJ: You used a model called the general-relativistic magnetohydrodynamic (GRMHD) simulation to predict the behaviors of the accretion disks and the mergers of neutron stars.² Could you briefly describe this model?

EQ: If you're near a neutron star or a black hole, you are moving so close to the speed of light and the gravity is so strong that Newton's theory of gravity and motion doesn't really apply. Einstein's theory of general relativity gives us a more complete model. The magnetohydrodynamic aspect of the model looks into how charged gases subjected to electric and magnetic forces behave. It's a theory for something similar to the atmosphere of the Earth, but in Earth's atmosphere, gas is mostly neutral, so it doesn't interact with electric and magnetic forces. However, if we want to describe a gas where these forces are important, we need a model that can measure that. This particular model tries to study how material would orbit around a black hole in a disk, how the material would get blown away, and how that could be observed.

BSJ: What are the differences between gravitational waves and electromagnetic waves, and how were you able to use their data to create the GRMHD model?

“The gravitational waves allowed us to see the collision itself, and the light we observed was from the materials that were flung off from the collision.”

EQ: All forms of light—radio waves, X-rays, gamma rays—are basically changes in the strength of electric and magnetic fields. These waves travel through space at the speed of light and carry information that we are able to observe at the right wavelength. Gravitational waves are something completely different. Small changes in the strength of gravity travel through space at the speed of light. Thanks to their inherently different properties, if you can see both electromagnetic and gravitational waves, you learn very different things about what’s going on in the object that produced them. Gravitational waves tell you about the mass, and light waves indicate more about the behavior of the material that was flung off. In the case of these colliding neutron stars, the gravitational waves allowed us to see the collision itself, and the light we observed was from the materials that were flung off from the collision. By probing different parts of the problem, we can see a much more complete story of what happened.

BSJ: Following the 2017 neutron star merger, how accurate were your predictions based on the GRMHD model?

EQ: This paper was actually a collection of computer simulations that took longer than expected. We started the simulations before the merger was detected, and

we obtained results after the detection. It turns out that the conditions we simulated were a pretty good match of what we observed. A lot of our initial results were in the ballpark of what we needed to explain our observations. We already knew that when two neutron stars would collide, they would collapse, create a black hole, and throw off a certain amount of mass into space. We had a rough sense of what those numbers were from previous theoretical calculations, so even though we did not know anything about the observations when we started doing the work, we knew roughly what the right thing to calculate was. Now that the observations are in hand and we know exactly what we want to explain, we can go back and make more refined calculations.

BSJ: What is the significance of being able to predict and analyze the behaviors of these cosmic events such as neutron star mergers?

EQ: The truth is these are hard problems, and a lot of the predictions I make do not turn out to be quite so right. You make approximations when you try to figure something out, so what was nice about this case is that most of the predictions were at least roughly right. I think this is of broad scientific interest because it is the first time in scientific history, at least on Earth, that we have seen the same object produce both gravitational waves and light. And that, as I have alluded to earlier, gives us very different information. Now we can learn a lot more about

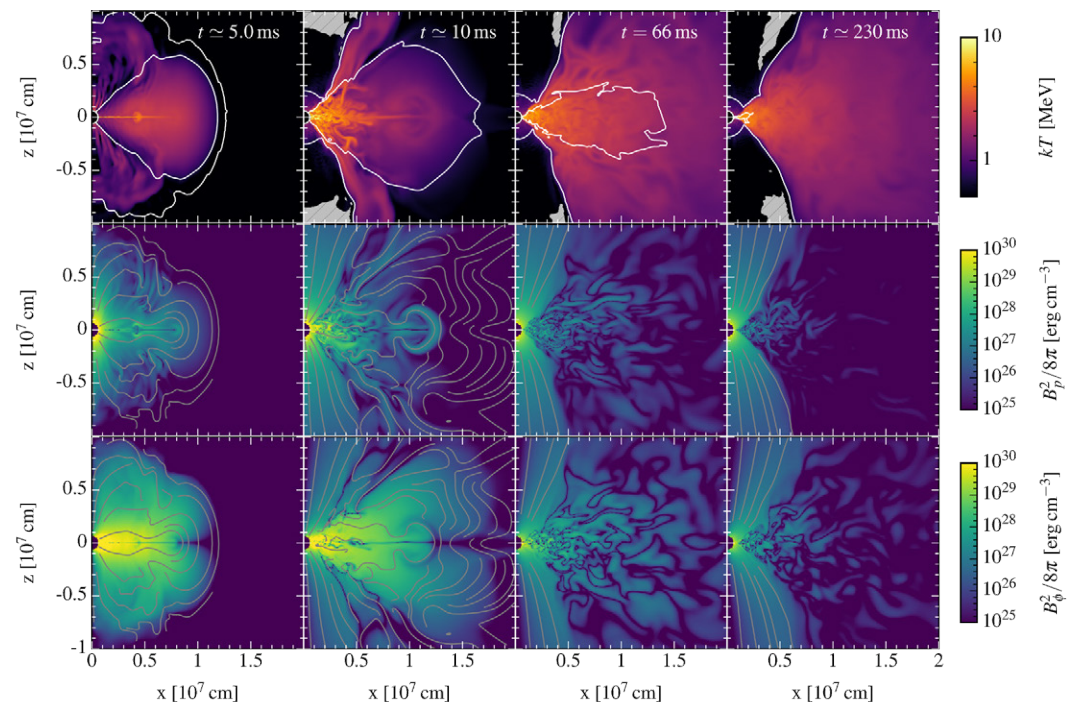


Figure 3: Evolution of the GRMHD model.² From top to bottom, each row represents temperature, poloidal magnetic pressure, and toroidal magnetic pressure at four different time points. Magnetic field lines are shown in gray.

what actually happened in the event by combining two different views of the same phenomenon. In addition, we know that there is gold, uranium, and platinum on Earth, but we did not know where in nature these elements actually came from. This event provided evidence that these elements are produced in colliding neutron stars. So, this solves a 60- or 70-year-old problem of identifying where in nature elements that exist on Earth are produced. Understanding the nature of matter, atoms, and protons and neutrons in the nucleus has been an essential problem in physics over the past few centuries. In astrophysics, it has been figuring out where hydrogen, carbon, and iron come from. This particular event helped complete our understanding of the story of where the basic building blocks of everything here on Earth comes from.

BSJ: What advice or activities would you recommend to anyone looking to enter into astrophysics?

EQ: I think there are two important things. One is taking basic physics classes to get a grounding in physics. Another is learning to be comfortable doing calculations with computers, in Python or something like that, because more and more of what we do is computer-based calculations. Even if you can figure something out like algebra and geometry with a pencil and paper, more and more often you have to build computer models of what you are trying to understand. Another thing I encourage students who might be interested in science to do is try their hand at undergraduate research. Research is very different from doing classwork. It is a lot more frustrating, usually, because you don't know the answer. The problem that you are trying to solve usually takes a much longer time. If a homework problem takes five hours, that is usually a long homework problem. In research, there were problems that I worked on that took more than a year to figure out. And frankly, dealing with that frustration is something that people either really like and it becomes motivating, or people get irritated and it is demoralizing. I think that figuring out how you approach that kind of work is very useful, even if you end up working in industry. If you work at a startup, there is a similar, long-term horizon to the kinds of problems that people work on.

“The 2017 neutron star merger is the first time in scientific history that we have seen the same object produce both gravitational waves and light.”

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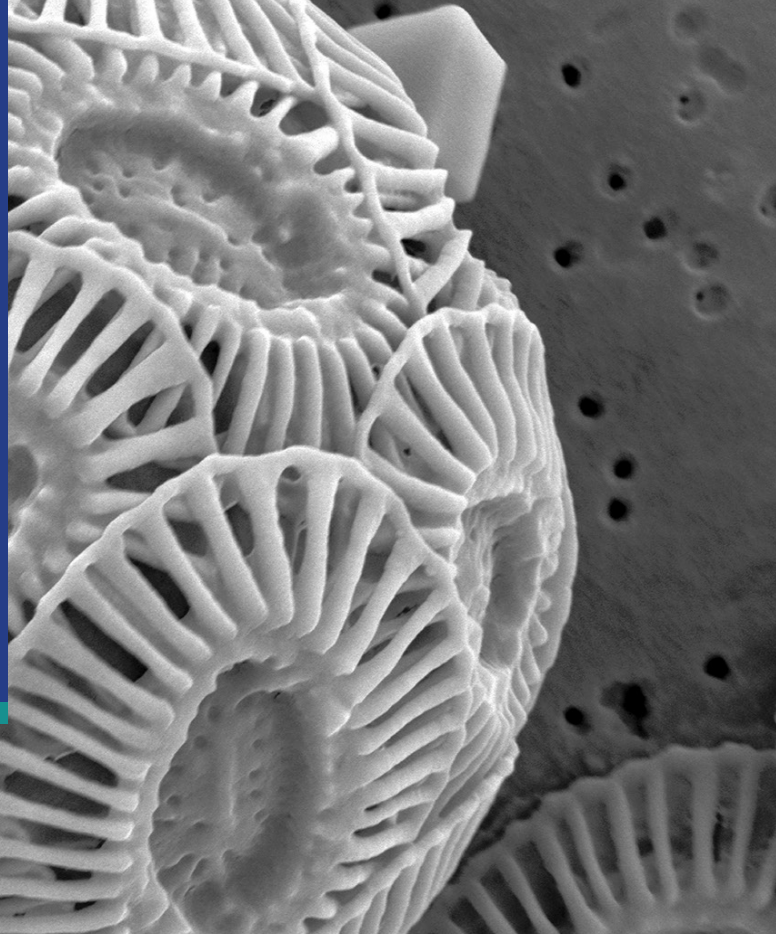
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THE BIOLOGICAL CARBON PUMP: CLIMATE CHANGE WARRIOR

BY MADALYN MILES



Imagine a cold, windy day out on the open ocean, and your robot just got chewed up by a shark.

For Jim Bishop, a professor of Marine Science at UC Berkeley, conducting research at sea comes with risks such as these. But a bigger challenge is the well-being of the biochemical mechanism that his team is studying beneath the waves. This biochemical mechanism is called the “Biological Carbon Pump,” and it may help calm the crisis of global warming. The Pump naturally sinks as much carbon to ocean depths as is found in the atmosphere, but as humans continue to emit greenhouse gases, can the Pump keep up?¹ Professor Bishop and PhD student Hannah Bourne are studying the Pump’s process of carbon sequestration, hoping that by understanding its pathways, they may be able to help the Pump help us.

THE BIOLOGICAL CARBON PUMP: THE BASICS

The Pump’s traditionally understood salt-shaker mechanism is relatively simple. Picture a vertical sequestration of carbon through a marine food chain. Phytoplankton, such as coccolithophores, take up carbon into their shells. Then, they are preyed on by larger organisms and sink to the bottom of the ocean as feces to be buried via deep-sea sedimentation. On a molecular level, carbon starts as carbon dioxide in the atmosphere. Most winds up in the bodies of phytoplankton through the movement of three dissolved organic compounds: carbonic acid (H_2CO_3), bicarbonate (HCO_3^-), and carbonate (CO_3^{2-}).² First, atmospheric carbon dioxide is absorbed in the upper euphotic layer of the ocean and mixes with

water molecules to form carbonic acid.³ Second, H_2CO_3 discards its hydrogen atoms one by one to become HCO_3^- , and then a CO_3^{2-} ion, which coccolithophores will use to form their calcium carbonate (CaCO_3) plates on their shells (Fig. 1a). Once these plated primary producers are consumed by larger organisms such as zooplankton (Fig. 1b-c), they become ballast-like fecal pellets that rain down like salt towards the deep sea (Fig. 1d).⁴

Scientists may reasonably understand this traditional salt-shaker mechanism, but understanding what controls the Pump’s rate of carbon flux is more complicated. Twenty years and six research papers later, Professor Bishop and his research team have been convinced that primary productivity in the ocean and the rate of carbon flux vary across space and time.

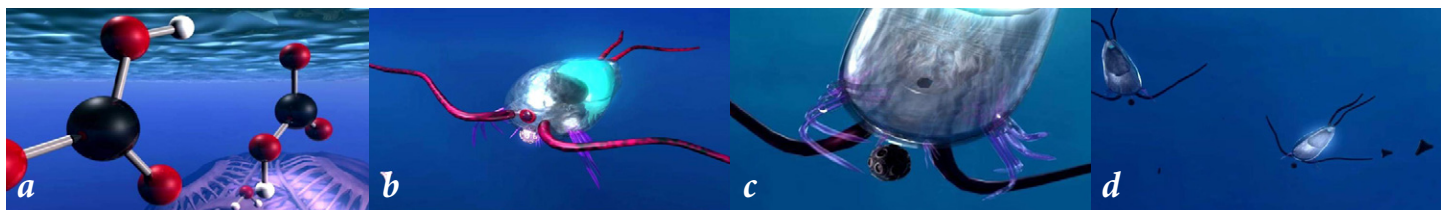


Figure 1: The steps of the ocean’s Biological Carbon Pump.



Figure 2: A CFE robot breaks the surface of the ocean as it is launched. It is ready to monitor how much carbon the Pump is sending downwards.

VARIATION ACROSS TIME

Professor Bishop observed the seasonal variations of primary productivity back in 1996. While at the Lawrence Berkeley National Lab, he launched the first set of ocean profiling robots called the Carbon Flux Explorers (CFEs) in the North Pacific, just off Canada's west coast, in order to measure the concentrations of carbon particles in the ocean. Within weeks, nature delivered a dust storm that swept from Asia across the North Pacific, dusting and fertilizing the sea with iron. Bishop, along with two other scientists from the Scripps Institution of Oceanography in La Jolla, California, hypothesized that an increase in iron, a limiting nutrient for phytoplankton, would stimulate primary productivity, and therefore carbon flux.^{5,6}

"Sure enough, we got a signature of the stimulation of the biology as a result of the deposition of dust iron," Professor Bishop says of the 1996 experiment. "But the effect only lasted for three weeks!"

"By understanding its pathways, [scientists] may be able to help the Pump help us."

The reason the effect was so short-lived is that the Pump's turnover time is lightning quick. While landlocked carbon cycles circulate on a time scale of over two dozen years; for example, the marine carbon cycle circulates in as few as two weeks, according to Professor Bishop.⁷ This rapid turnover time is due to the radically different growth cycles of phytoplankton and land plants—many coccolithophores live for merely a week, according to Professor Bishop, before being absorbed into the food chain and excreted as ballast-like fecal pellets, while land plants have far less turnover. "You cannot go out on ships every three months and do a seasonal study on this ocean biological pump, because that is equivalent to sitting here in my office at UC Berkeley, closing the blinds, leaving them closed, and opening them once every 240 years," Jim says.

VARIATION ACROSS SPACE

This reality drove Professor Bishop and colleagues to deploy their CFEs again the following year, this time while studying the Pump in the Southern Ocean.⁸ The team knew that diatoms, another important phytoplankton that accounts for nearly 40% of marine primary productivity, prefer cold water.⁹ Their hypothesis was that iron fertilization would have a minimal effect in the higher latitude control group while a "massive sedimentation event" would oc-

cur in the colder, lower latitude, waters—a diatom's paradise.

But the results were unexpected. "To the surprise of everyone, the place that was not supposed to have a response to iron, did!" says Professor Bishop. This clued marine scientists into the possibility that primary productivity is not the only predictor of the Pump's success.

Fifteen years later, after observing an phytoplankton bloom in the California Current during summer of 2017, Bourne made her own observation that, while chlorophyll—a sign of primary productivity—was over 30 times higher in concentration closer to shore, carbon flux rate was the same onshore as in the deeper, offshore location of the plume.

ZEROING IN ON AN EXPLANATION WITH MULTIPLE PATHWAYS

Professor Bishop and his colleagues made two overarching observations from their respective research voyages. First, they observed that enhancing primary productivity by raining down limiting nutrients, like iron, over the euphotic zone does not guarantee carbon flux increase. Second, they observed that temperature cannot necessarily predict primary productivity and flux rate either. Thus, the traditional salt-shaker pathway cannot be the Biological Carbon Pump's only mechanism. "We are actually finding out that this biological

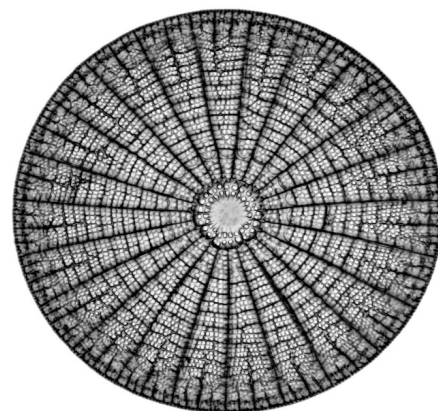


Figure 3: Diatom, a phytoplankton that can play an important role in the Biological Carbon Pump.



Figure 4: Undergraduate Sylvia Targ, PhD student Hannah Bourne, and Professor Jim Bishop.

carbon pump is just not simply a monolithic process,” Professor Bishop explains, “and there are multiple [biological] pathways by which carbon sinking can be enhanced.”

As part of her PhD thesis, Hannah Bourne has come up with two hypotheses that explain the following observation: more phytoplankton present in the euphotic zone does not necessarily mean increased carbon flux. One is that an active organism can swim from depth to the surface, feed, and then swim back down and excrete feces. The other is that sediments from shallow shelves can flow into the interior, and the filter-feeders will then harvest the material. The latter scenario would explain the high fluxing at depths that Bourne observed in some locations in the California Current.

Bourne and Professor Bishop still have much to discover. But by actually observing carbon aggregates, or particulate organic carbon, at different places, they are piecing together a puzzle of how the Pump works. “It is a bit like how by going through garbage cans you can tell how people live,” Jim says. “We can actually see how [respective] ecosystems are functioning to transport carbon.”

To Bourne, this project is very exciting. “What I really love about our research is we

put these robotic instruments out, and then they come back with large series of images of what is sinking down through the water column,” says Bourne. “It is completely different [from the] things I see in my daily life back on land, and it is cool to see this completely different part of the Earth,” she says.

HORIZONS FOR FUTURE EXPLORATION

What is next for these UC Berkeley marine scientists studying the Pump? “I think we should learn as much as we can now so we can predict how it might change as climate is changing,” Bourne says.

And they are getting closer. Bourne and Professor Bishop’s summer 2017 voyage revealed that the robots are even more effective at calculating carbon flux than the team had supposed, which makes Professor Bishop all the more eager to see them in action at full capacity.¹⁰

Next, Professor Bishop is anxious to start observing these changes on the proper time and space scale to eventually give the public a full picture of the Pump’s pathways. He believes that proactive use of these robots is the key to unlocking the secrets of the Biological Pump’s variations across time and space. “We have three robots and there is a big ocean out there!” Bishop says. “Let’s put thirty or forty of them out in the California Current in an upwelling system and let them go! Let’s go to area where we did SOFeX and let them go south of Tasmania, and they’ll go a third of the way around the world in a year!”

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